The ASM Lab ventilation solver

How the airflow solver, the linear algebra and the coupled airflow–thermal solver work, with worked examples computed step by step

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Introduction

This document explains how ASM Lab computes airflow and air conditions in a mine ventilation network. It is written to be read alongside the program: every equation is the one the program uses, and every worked example is taken from the program's own test networks, with the numbers recorded from the solver as it ran. Where a calculation is shown by hand, the same figures can be reproduced on a calculator or in a spreadsheet.

Part A covers the airflow solver: the airway laws (Atkinson resistance, density, natural ventilation, fans, regulators), the nodal formulation of Kirchhoff's laws, Newton–Raphson iteration, the Jacobian, line search and convergence. It then follows the solver iteration by iteration through four networks: parallel drives, a fan on its curve, a Wheatstone bridge and a natural-ventilation loop.

Part B covers the linear systems solved at every iteration: Gaussian elimination and LU, the Cholesky factorisation, sparse envelope storage and the reverse Cuthill–McKee ordering, conjugate gradients with incomplete Cholesky, and, for the non-symmetric coupled system, the LU + GMRES combination and why GMRES is used.

Part C covers the coupled solver, in which pressures, enthalpies and moistures are solved together in one Newton system: the thermal model (autocompression, Gibson rock heat, latent heat, mixing, still air), the coupled equations and Jacobian, and a full worked example with hand checks of the heat transfer, the natural ventilation pressure and the energy balance.

Units are SI throughout: pressure in Pa, flow in m^3/s, mass flow in kg/s, enthalpy in kJ per kg of dry air, moisture in g or kg of water per kg of dry air, resistance in $N \cdot s^2/m^8$. Gravity is 9.79 m/s^2, following McPherson's field book and VUMA.

Part A — The airflow solver

A1. What the solver computes

A mine ventilation network is a set of *airways* (drives, shafts, raises, crosscuts, ducts) joined at *nodes* (junctions, shaft stations, portals). Air moves through it because something creates a pressure difference: a fan, the weight difference between a cold downcast and a warm upcast column of air (natural ventilation), or a pressure imposed at the surface connections. The solver's job is to find, for a given network, the steady state in which every airway carries the flow its pressure difference drives and no node gains or loses air.

The result is the **pressure at every node** and the **volume flow, mass flow and pressure drop in every airway**, plus the fan operating points, regulator settings and the air density everywhere. Everything else the program shows (velocities, power, colour maps, compliance checks) is derived from these.

ASM Lab solves the network by the **nodal pressure method** with **Newton–Raphson** iteration: the unknowns are the pressures at the free nodes, and the equations are conservation of mass at each of those nodes. This part explains each ingredient, from the airway law to convergence, and then follows the solver step by step through four small networks, with every number written out so it can be checked by hand.

Notation

Table 1. Notation used in Part A.

Symbol	Meaning	Unit
P_i	absolute static pressure at node i	Pa
q_e	volume flow in airway e, positive from its source node to its target node	m³/s
ρ_e	air density in airway e	kg/m³
R_e	Atkinson resistance of airway e at its operating density	N·s²/m⁸
z_i	elevation of node i	m
Δ_e	effective pressure difference driving airway e	Pa
F_i	mass residual (imbalance) at free node i	kg/s
J	Jacobian matrix, J_ij = ∂F_i/∂P_j	kg/(s·Pa)
g	gravitational acceleration, 9.79 (McPherson; VUMA)	m/s²

A2. The network as a graph

The network is held as a directed graph. Each airway has a *source* and a *target* node; the direction only fixes the sign convention (a negative flow simply runs the other way). Nodes are of two kinds:

- **Boundary (fixed-pressure) nodes**: surface intakes and exhausts, portals, or any node given a pressure. Their pressure is known and never changes.

- **Free nodes**: every other junction. Their pressure is unknown; there is one equation (mass balance) for each.

Figure 1 shows the Wheatstone bridge test case used later (lit2): four nodes, five airways, one intake A and one exhaust D. B and C are free nodes; the diagonal BC connects the two sides.

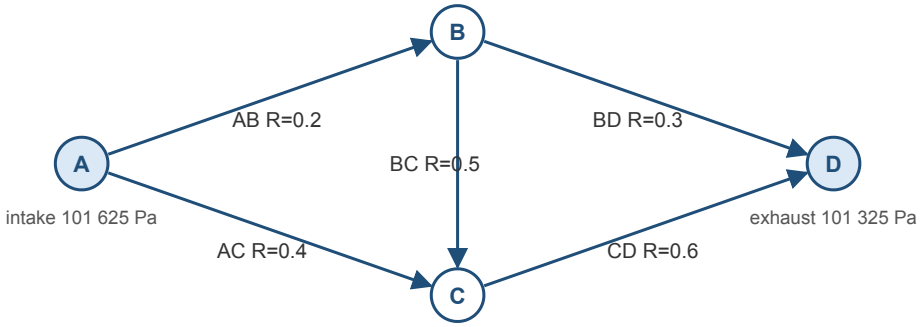


Figure 1. The balanced Wheatstone bridge (lit2). Shaded: fixed-pressure nodes. Arrows show each airway's source → target direction.

Dead ends. A free node with only one airway can balance its mass only at zero flow, so that airway carries nothing. Before solving, ASM Lab peels such stubs off repeatedly (removing one may expose the next along the same spur) and reports them at zero flow. An airway with a fan or regulator on it is never removed silently: a device on a dead end is a modelling error the user must see. Pruning keeps the linear system smaller and stops the tiny, noisy imbalance of a stub from deciding convergence.

A3. Airway resistance

The pressure lost to friction in an airway follows the **square law** of Atkinson:

$$\Delta P = R\,Q\,|Q| \tag{A1}$$

The $Q|Q|$ form keeps the sign: air flowing backwards loses pressure in the backward direction. From the geometry, at the reference density $\rho_{\mathrm{ref}} = 1.2$ kg/m³,

$$R_{\mathrm{std}} = \frac{k\,L\,\mathrm{Per}}{A^3} + \frac{X\,\rho_{\mathrm{ref}}}{2A^2} \tag{A2}$$

where k is the Atkinson friction factor (kg/m³), L the length, Per the perimeter, A the cross-sectional area and X the dimensionless shock-loss factor (bends, junctions, obstructions: $\Delta P_{\mathrm{shock}} = X\rho V^2/2$). Rectangular, arched (semicircular roof) and circular profiles are supported, and a stated area and perimeter (an irregular VUMA profile) override the shape formulas. A user can also give R directly (a resistance override).

Worked example (case02). A 5 m × 5 m drive, 100 m long, $k = 0.01$ kg/m³, no shock loss: $A = 25$ m², $\mathrm{Per} = 20$ m, so

$$R_{\mathrm{std}} = \frac{0.01 \times 100 \times 20}{25^3} = \frac{20}{15\,625} = 1.280 \times 10^{-3}\ \mathrm{N\,s^2/m^8}$$

Minimum resistance. A zero resistance is not a low-loss airway but a degenerate one: its two ends are the same point and the pressure difference across it is undefined. Every R is floored at $R_{\min} = 10^{-6}$ N·s²/m⁸, which at 100 m³/s costs only 0.01 Pa. (MFIRE and VUMA write 0 on fan branches as a convention; the floor reads it as “negligible”.)

Density scaling. R as given is for standard air. Friction and shock losses scale with density, so the solver uses $R = R_{\mathrm{std}}\,\rho_e/\rho_{\mathrm{ref}}$ (unless the resistance is locked). In case02 the airway density is 1.19950 kg/m³ (Section A4), giving $R = 1.280 \times 10^{-3} \times 1.19950/1.2 = 1.2795 \times 10^{-3}$, exactly the value in the solver's trace (Section A13).

Regulators are modelled as an extra resistance from their opening area A_r (Blackbook eq. 2.18; McPherson $A_r = 1.2\,Q\sqrt{\rho/\Delta P}$):

$$R_{\mathrm{reg}} = \frac{1.44\,\rho}{A_r^2} \tag{A3}$$

A regulator set to a target pressure drop has its area found from the flow it carries; a fixed-quantity airway is treated as a current source (its flow is prescribed and does not depend on pressures).

A4. Air density

Mass, not volume, is conserved at a junction, so the solver needs the density of the air in every airway. It is computed from the pressure, the dry-bulb temperature and the moisture at each node, using the psychrometric relations of the program:

$$e_s(t) = 610.5\,\exp\!\left(\frac{17.27\,t}{237.3 + t}\right)\ \mathrm{Pa} \quad (\text{Tetens; Blackbook eq. } 5.6) \tag{A4}$$

$$p_w = e_s(t_{wb}) - \gamma\,P\,(t_{db} - t_{wb}), \qquad \gamma = 6.44 \times 10^{-4}\ \mathrm{K^{-1}} \tag{A5}$$

$$\rho = \frac{P - 0.378\,p_w}{287.04\,(t_{db} + 273.15)} \tag{A6}$$

where p_w is the partial pressure of water vapour (from the wet bulb by the psychrometer equation, or from the relative humidity, $p_w = \mathrm{RH} \cdot e_s(t_{db})$). Moist air is lighter than dry air at the same pressure and temperature, which the $0.378\,p_w$ term accounts for. A node without a stated climate uses 20 °C and 50 % RH.

Worked example (case02). Node A is at 101 425 Pa and node B at 101 325 Pa, both 20 °C and 50 % RH. $e_s(20) = 610.5\,e^{17.27\times20/257.3} = 2337.1$ Pa, so $p_w = 0.5 \times 2337.1 = 1168.6$ Pa, and

$$\rho_A = \frac{101\,425 - 0.378 \times 1168.6}{287.04 \times 293.15} = 1.20010, \qquad \rho_B = \frac{101\,325 - 0.378 \times 1168.6}{287.04 \times 293.15} = 1.19891\ \mathrm{k}$$

The airway density is the mean of its two end nodes: $\rho_e = (1.20010 + 1.19891)/2 = 1.19951$ kg/m³ — the solver's value is 1.19950 (the small difference comes from solving the wet bulb back from the humidity and recomputing p_w by Eq. A5).

Because density depends on pressure, and pressure is the unknown, the densities are **refreshed at the start of every Newton iteration** and again after each accepted step (so convergence is judged on the state that will be reported). After the solve the program reports the density spread; above 5 % the air can no longer be treated as having one density and a warning recommends running with real node temperatures (the coupled solver of Part C).

A5. Elevation and natural ventilation

A column of air has weight. Between two nodes at different elevations, the pressure difference that drives flow is not $P_s - P_t$ but that difference corrected for the weight of the air column between them:

$$\Delta_e = P_s - P_t - \rho_e \, g \, (z_t - z_s) \tag{A7}$$

For a downcast shaft ($z_t < z_s$) the correction is positive: air flows down a shaft even with equal pressures at both ends if nothing resists it, and the pressure at the bottom is higher by $\rho g H$. Natural ventilation is the consequence: in a closed loop with a cold downcast and a warm upcast, $\sum \rho_e g \, \Delta z_e$ around the loop does not cancel, and the difference, the **natural ventilation pressure**,

$$\mathrm{NVP} = g \, H \, (\bar{\rho}_{\mathrm{down}} - \bar{\rho}_{\mathrm{up}}) \tag{A8}$$

drives air round the loop like a fan. In the solver no special term is needed: Eq. A7 applied to every airway, with each airway's own density, produces it automatically. Section A16 (case09) shows it.

A6. Fans

A fan is an airway that adds pressure. Its law combines the airway's own resistance with the fan characteristic $P_{\mathrm{fan}}(q)$:

$$R \, q \, |q| - P_{\mathrm{fan}}(q) = \Delta_e \tag{A9}$$

Two forms of characteristic are used:

- **Quadratic:** $P_{\mathrm{fan}} = a - b \, q|q|$. Then Eq. A9 becomes $(R + b) \, q|q| = \Delta_e + a$ and is solved in closed form like an ordinary airway with $R_{\mathrm{eff}} = R + b$.
- **Tabulated curve** (manufacturer points, linearly interpolated, scaled by ρ/ρ_{ref}): on each segment $P_{\mathrm{fan}} = Mq + B$, so Eq. A9 is the quadratic $Rq^2 - Mq - (B + \Delta_e) = 0$. Every segment's roots are computed exactly and the root closest to the previous iterate is kept, which avoids the fan jumping between operating points (and keeps it out of the stall zone where possible).

The fan's slope dP_{fan}/dq enters the Jacobian (Section A9), so a steep fan curve makes the fan's flow insensitive to the network pressures, exactly as in reality.

A7. The equations: Kirchhoff's laws for air

Two laws govern a network, the ventilation equivalents of Kirchhoff's laws for electrical circuits:

1. **Mass conservation at a node** (Kirchhoff's current law): the mass of air entering a node equals the mass leaving it, $\sum_{\text{in}} \rho_e q_e = \sum_{\text{out}} \rho_e q_e$.
2. **Pressure around a loop** (Kirchhoff's voltage law): around any closed loop the pressure drops, fan pressures and natural ventilation sum to zero.

Classical hand methods (Hardy Cross) take the airway flows as unknowns, satisfy law 1 by construction and iterate on law 2 loop by loop. ASM Lab instead takes the **node pressures** as unknowns. Then law 2 holds automatically: any loop's pressure drops are differences of node pressures, which telescope to zero. Only law 1 remains, one equation per free node:

$$F_i(P) = \sum_{e:\, t(e)=i} \rho_e\, q_e(P) - \sum_{e:\, s(e)=i} \rho_e\, q_e(P) = 0, \qquad i \in \text{free nodes} \tag{A10}$$

with each airway's flow given by its own law, inverted:

$$q_e = \operatorname{sign}(\Delta_e) \sqrt{\frac{|\Delta_e|}{R_e}} \tag{A11}$$

The nodal form has practical advantages over loop methods: no loop set has to be chosen (a mine with thousands of airways has thousands of independent loops), fixed-pressure boundaries at many surface connections are natural, and the linear systems are symmetric and sparse, which Part B exploits. Its one difficulty is visible in Eq. A11: near zero flow the square root is infinitely steep. Section A10 explains how the solver deals with that.

A8. Newton–Raphson iteration

Eq. A10 is a system of nonlinear equations $F(P) = 0$. Newton's method replaces it, at the current estimate $P^{(k)}$, by its linearisation

$$F(P^{(k)} + \delta P) \approx F(P^{(k)}) + J(P^{(k)})\, \delta P \tag{A12}$$

and chooses the step that makes the linearisation zero:

$$J(P^{(k)})\, \delta P = -F(P^{(k)}), \qquad P^{(k+1)} = P^{(k)} + \alpha\, \delta P \tag{A13}$$

where $\alpha \le 1$ is chosen by a line search (Section A11). Near the solution Newton's method converges **quadratically**: the number of correct digits roughly doubles each iteration. This is visible in every example below: the residual falls slowly while the estimate is far away, then drops by orders of magnitude in the last two or three iterations.

One-dimensional illustration

For a single free node (case05, Section A14) the method is the familiar tangent construction: $\delta P = -F/F'$. Each step follows the tangent of the mass-imbalance curve $F(P)$ to where it crosses zero.

A9. The Jacobian

Differentiating Eq. A10, only airways touching node i contribute to row i. For airway e from s to t, $\partial\Delta_e/\partial P_s = 1$ and $\partial\Delta_e/\partial P_t = -1$, and from Eq. A9 by implicit differentiation,

$$\frac{\partial q_e}{\partial \Delta_e} = s_e = \frac{1}{|\,2R_e|q_e| - dP_{\text{fan}}/dq\,|} \tag{A14}$$

(for a plain airway simply $1/(2R|q|)$). With densities held at their values for the iteration, the entries are

$$J_{ii} = -\sum_{e \ni i} \rho_e s_e, \qquad J_{ij} = \sum_{e:\, i \leftrightarrow j} \rho_e s_e \quad (i \neq j) \tag{A15}$$

Airways to a fixed-pressure node contribute to the diagonal only, since the boundary pressure is not an unknown. Three properties follow, and Part B relies on all of them:

- **Symmetric:** $J_{ij} = J_{ji}$, since the same airway gives both entries.
- **Negative definite (diagonally dominant):** each diagonal is minus the sum of the positive slopes at the node, so $|J_{ii}| \geq \sum_{j \neq i} J_{ij}$, with strict inequality at any node connected to a boundary. Hence $B = -J$ is a symmetric positive definite M-matrix.
- **Sparse:** row i has nonzeros only for the neighbours of i, typically 3 or 4 in a mine.

A **fixed-flow airway** (a fixed-quantity regulator, or a booster modelled as a flow) has $s_e = 0$: its flow does not depend on pressures, so it adds its prescribed ρq to the residual but nothing to the Jacobian, exactly as a current source in circuit analysis.

A10. The zero-flow singularity and the slope cap

Eq. A14 has a problem at $q = 0$: $s = 1/(2R|q|)$ is infinite. Physically, the square-root law $q = \sqrt{\Delta/R}$ has a vertical tangent at $\Delta = 0$. The solver caps the denominator at 10^{-6}, so $s \leq 10^{6}$ m³/s per Pa. This happens in two situations:

- **At the start:** if the initial pressures make two connected nodes equal, the airway between them has zero flow and a slope of 10^6. The Newton step is then tiny (Section A15, iteration 1). After one or two small steps the flows become nonzero, the slopes realistic, and Newton proceeds normally.
- **At the solution:** an airway whose true flow is (nearly) zero, like the diagonal of a balanced bridge. Here a full Newton step from $+\Delta$ lands on $-\Delta$: for $q = \sqrt{\Delta/R}$ the secant across zero is exactly twice the tangent, so the iterate jumps to the mirror image

and back. The Armijo line search (next section) prevents this period-2 cycle by requiring a real decrease, which forces a half step that lands inside the true solution's basin.

A11. Line search, tolerance and stopping

Armijo line search. A full Newton step can overshoot when the estimate is far from the solution. The solver first tries $\alpha = 1$; it accepts the step if the largest nodal residual falls by a sufficient amount,

$$\max_i |F_i(P + \alpha\,\delta P)| \le (1 - c\,\alpha)\,\max_i |F_i(P)|, \qquad c = 0.01 \tag{A16}$$

and otherwise halves α, up to 10 times ($\alpha \approx 0.002$). If no trial passes, the last (smallest) step is taken anyway, which guarantees progress through the zero-flow situation of the previous section. Requiring a *sufficient* decrease rather than any decrease is what breaks the period-2 cycle: on a 200-rung ladder test the old “accept if not worse” rule ran 2000 iterations without converging; the Armijo rule converges in 14.

Tolerance. The solve has converged when the largest nodal mass imbalance is below

$$\mathrm{tol} = \max\!\left(10^{-4}\ \mathrm{kg/s},\ 10^{-6} \times \max_e |\rho_e q_e|\right) \tag{A17}$$

i.e. one part in a million of the largest mass flow in the network, but never tighter than 0.1 g/s. A fixed absolute tolerance fails both ways: it asks a 600 kg/s mine for one part in six million (close to the round-off floor of the summation itself) while letting a small network pass with a badly wrong flow. The tolerance is re-evaluated every iteration, because the first estimate can carry flows far larger than the solution.

Stall and precision limit. The solver also stops if the residual has not improved by 0.1 % in 40 iterations (it then reports the node that dominated the residual and whether a fan sits on it, which usually identifies a fan curve that cannot span the pressure swings imposed on it), or if the residual has not changed by one part in 10^9 for 5 iterations while within 10× the tolerance. The latter is reported as convergence at the precision limit of double-precision arithmetic: the Newton step no longer moves the pressures. The hard cap is 2000 iterations.

Initial estimate. Free nodes start at the first boundary pressure plus the weight of air down to their elevation, $P_i^{(0)} = P_{\mathrm{ref}} + \rho_0 g\,(z_{\mathrm{ref}} - z_i)$, so deep nodes do not start at surface pressure. A previous solution is used instead when available (a warm start, e.g. after a small edit).

A12. Summary of one iteration

1. Refresh densities: $\rho_i(P_i, T_i, \mathrm{moisture})$ at every node (Eq. A6); ρ_e = mean of the end nodes; $R_e = R_{\mathrm{std}}\,\rho_e/\rho_{\mathrm{ref}}$.
2. For every airway: Δ_e (Eq. A7), q_e (Eq. A11 or the fan root), slope s_e (Eq. A14, capped).
3. For every free node: residual F_i (Eq. A10) and Jacobian row (Eq. A15).

4. Solve $J\,\delta P = -F$ (Part B: envelope Cholesky, or preconditioned conjugate gradients for very large networks).
5. Line search on α (Eq. A16); update $P \leftarrow P + \alpha\,\delta P$.
6. Refresh densities at the new pressures; measure $\max|F|$; compare with the tolerance (Eq. A17); check stall and precision limit.

The following sections run this procedure on four test networks. All numbers come from the solver itself (an observation hook records every quantity of every iteration), and each step is shown in enough detail to reproduce by hand or in a spreadsheet.

A13. Example 1 — two parallel drives (case02)

Two identical 5 m × 5 m drives, 100 m long, connect intake A (101 425 Pa) to exhaust B (101 325 Pa). Both nodes are fixed, so there is no free node and nothing to iterate: every airway's flow follows directly from Eq. A11. The expected answer (McPherson, parallel airways) is 279.508 m³/s each at standard density.

$$\Delta = 101\,425 - 101\,325 = 100\ \mathrm{Pa}, \qquad q = \sqrt{\frac{100}{1.2795 \times 10^{-3}}} = 279.566\ \mathrm{m^3/s}$$

each, 559.132 m³/s in total, with mass flow $1.19951 \times 279.566 = 335.341$ kg/s per drive. The 0.02 % difference from the textbook 279.508 m³/s is the density correction: the air here is 1.1995 kg/m³, not 1.2, so R is 0.04 % lower and the flow 0.02 % higher ($279.508 \times \sqrt{1.2/1.1995} = 279.566$).

Table 2. case02: result.

Airway	R (Ns²/m⁸)	ρ (kg/m³)	q (m³/s)	ρq (kg/s)	Expected (m³/s)
t1	1.2795×10⁻³	1.19951	279.566	335.341	279.508
t2	1.2795×10⁻³	1.19951	279.566	335.341	279.508

A14. Example 2 — a fan on a curve (case05)

A fan with characteristic $P = 2000 - 0.2Q^2$ (tabulated every 2 m³/s from 0 to 100 m³/s) sits in airway A→M, and a system resistance $R = 0.5$ in M→B; A and B are both at 101 325 Pa. The analytic operating point is where fan pressure equals system loss: $2000 - 0.2Q^2 = 0.5Q^2$, so $Q = \sqrt{2000/0.7} = 53.452$ m³/s and $P_{\mathrm{fan}} = 1428.6$ Pa at standard density.

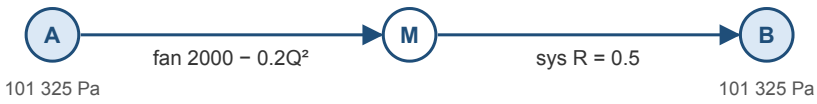


Figure 2. case05: one free node, M.

M is the only free node, so J is a 1×1 matrix and each Newton step is a division. The table below lists all eight iterations.

Table 3. case05: every Newton iteration.

Iter.	P_M before (Pa)	max |F| before (kg/s)	α taken	max |F| after (kg/s)	Tolerance (kg/s)
1	101325.0000	119.89	0.001953	119.89	1.199×10^{-4}
2	101325.0000	119.89	0.062500	119.78	1.199×10^{-4}
3	101325.0039	119.78	1.000000	114.60	1.196×10^{-4}
4	101333.7970	114.60	1.000000	75.657	1.084×10^{-4}
5	101696.2933	75.657	1.000000	5.8307	1.000×10^{-4}
6	102678.2873	5.8307	1.000000	0.13701	1.000×10^{-4}
7	102755.1781	0.13701	1.000000	6.1656×10^{-4}	1.000×10^{-4}
8	102753.4477	6.1656×10^{-4}	1.000000	2.7176×10^{-6}	1.000×10^{-4}

Iterations 1–2: the zero-flow start. M starts at 101 325 Pa, equal to B, so the system airway M→B carries no flow and its slope is capped at 10^6. The Jacobian is dominated by that cap ($J = -1.1989 \times 10^6$) and the step is a mere 10.000×10^{-5} Pa. In iteration 1 no trial passes the Armijo test, so the smallest step is taken; in iteration 2 a sixteenth of the step passes. Even these tiny pressure rises start air moving through M→B, its slope drops to a realistic value, and from iteration 3 Newton takes full steps.

Iteration 3 in full:

Starting point of iteration 3. The node pressures entering this iteration are listed below; the boundary pressures never change.

Table 4. case05, iteration 3: node pressures before the step.

Node	Role	Pressure P (Pa)
A	fixed (boundary)	101325.0000
M	free (unknown)	101325.0039
B	fixed (boundary)	101325.0000

Step 1: the flow in every airway. For each airway the effective pressure difference is $\Delta = P_s - P_t - \rho g\,(z_t - z_s)$, and the flow follows from the inverted square law $q = \mathrm{sign}(\Delta)\sqrt{|\Delta|/R}$. The slope $s = \partial q/\partial\Delta = 1/(2R|q|)$ is what the Jacobian needs.

Table 5. case05, iteration 3: airway flows and slopes.

Airway	s→t	$P_s - P_t$ (Pa)	ρgΔz (Pa)	Δ (Pa)	R (Ns²/m⁸)	q (m³/s)	ρ (kg/m³)	ρq (kg/s)	s (m³/s per Pa)
fan	A→M	−0.0039	0.000	−0.0039	1.0000×10^{-6}	99.99965	1.19891	119.8907	0.048835
sys	M→B	0.0039	0.000	0.0039	0.50000	0.08839	1.19891	0.1060	11.314

Airway **fan** carries a fan, so its flow is the root of $R\,q|q| - P_{\mathrm{fan}}(q) = \Delta$ on the tabulated curve: $q = 99.99965$ m³/s, where the curve gives $P_{\mathrm{fan}} = 0.014$ Pa. Check: $1.000 \times 10^{-6} \times 99.9996^2 - 0.014 = -0.004$ Pa, which equals $\Delta = -0.004$ Pa.

For example, airway **sys**: $\Delta = 101325.004 - 101325.000 - (0.000) = 0.0039$ Pa, so $q = \sqrt{0.0039/0.50000} = 0.08839$ m³/s, and $s = 1/(2 \times 0.50000 \times 0.0884) = 11.314$.

Step 2: the mass residual at every free node. $F_i = \sum_{\text{in}} \rho_e q_e - \sum_{\text{out}} \rho_e q_e$ (kg/s), with "in" meaning airways whose target is node i.

$$F_M = 1.19891 \times (99.9996) - 1.19891 \times (0.0884) = 119.78469\ \text{kg/s}$$

Step 3: the Jacobian. Raising P_i by 1 Pa raises Δ on every airway leaving i and lowers it on every airway entering i, so $J_{ii} = -\sum_e \rho_e s_e$ over the airways at i, and $J_{ij} = +\rho_e s_e$ for an airway between i and j.

$$J_{MM} = -(1.19891 \times 0.048835 + 1.19891 \times 11.314) = -13.6225$$

Step 4: the Newton step. Solve $J\,\delta P = -F$.

$$\delta P_M = -\frac{F}{J} = -\frac{119.78469}{-13.6225} = 8.79312\ \text{Pa}$$

Step 5: the line search. The full step is tried first. It is accepted if the largest nodal residual falls to at most $(1 - 0.01\,\alpha)$ times its value before the step ($\max|F| = 119.78$ kg/s), otherwise α is halved.

Table 6. case05, iteration 3: line search.

Trial	α	P after trial	max |F| after trial (kg/s)	Required ≤ (kg/s)	Result
1	1.000000	M: 101333.7970	114.60	118.59	accepted

After the step the densities are refreshed at the new pressures and the residual is measured again: $\max|F| = 114.60$ kg/s against a tolerance of 1.1963×10^{-4} kg/s, so the solver continues.

Iteration 3 is still far from the answer (the fan delivers nearly 100 m³/s into a node that lets out almost nothing) but the step is in the right direction. The residual falls 114.6 → 75.7 → 5.83 → 0.137 → 6.2×10⁻⁴ → 2.7×10⁻⁶ kg/s: the last steps show the quadratic convergence. The final iteration in full:

Starting point of iteration 8. The node pressures entering this iteration are listed below; the boundary pressures never change.

Table 7. case05, iteration 8: node pressures before the step.

Node	Role	Pressure P (Pa)
A	fixed (boundary)	101325.0000
M	free (unknown)	102753.4477
B	fixed (boundary)	101325.0000

Step 1: the flow in every airway. For each airway the effective pressure difference is $\Delta = P_s - P_t - \rho g\,(z_t - z_s)$, and the flow follows from the inverted square law $q = \text{sign}(\Delta)\sqrt{|\Delta|/R}$. The slope $s = \partial q/\partial\Delta = 1/(2R|q|)$ is what the Jacobian needs.

Table 8. case05, iteration 8: airway flows and slopes.

Airway	s→t	P_s − P_t (Pa)	ρgΔz (Pa)	Δ (Pa)	R (Ns²/m⁸)	q (m³/s)	ρ (kg/m³)	ρq (kg/s)	s (m³/s per Pa)
fan	A→M	−1428.4477	0.000	−1428.4477	1.0000×10⁻⁶	53.45044	1.20740	64.5360	0.046881
sys	M→B	1428.4477	0.000	1428.4477	0.50000	53.44993	1.20740	64.5354	0.018709

Airway **fan** carries a fan, so its flow is the root of $R\,q|q| - P_{\text{fan}}(q) = \Delta$ on the tabulated curve: $q = 53.45044$ m³/s, where the curve gives $P_{\text{fan}} = 1437.258$ Pa. Check: $1.000 \times 10^{-6} \times 53.4504^2 - 1437.258 = -1437.255$ Pa, which equals $\Delta = -1428.448$ Pa. On this part of the curve the fan's slope is the 52–54 m³/s segment's, −21.2 Pa per m³/s at standard density, scaled by ρ/1.2.

For example, airway **sys**: $\Delta = 102753.448 - 101325.000 - (0.000) = 1428.4477$ Pa, so $q = \sqrt{1428.4477/0.50000} = 53.44993$ m³/s, and $s = 1/(2 \times 0.50000 \times 53.4499) = 0.018709$.

Step 2: the mass residual at every free node. $F_i = \sum_{\text{in}} \rho_e q_e - \sum_{\text{out}} \rho_e q_e$ (kg/s), with “in” meaning airways whose target is node i.

$$F_M = 1.20740 \times (53.4504) - 1.20740 \times (53.4499) = 0.00062\,\text{kg/s}$$

Step 3: the Jacobian. Raising P_i by 1 Pa raises Δ on every airway leaving i and lowers it on every airway entering i, so $J_{ii} = -\sum_e \rho_e s_e$ over the airways at i, and $J_{ij} = +\rho_e s_e$ for an airway between i and j.

$$J_{MM} = -(1.20740 \times 0.046881 + 1.20740 \times 0.018709) = -0.0791928$$

Step 4: the Newton step. Solve $J\,\delta P = -F$.

$$\delta P_M = -\frac{F}{J} = -\frac{0.00062}{-0.0791928} = 7.78552 \times 10^{-3}\ \text{Pa}$$

Step 5: the line search. The full step is tried first. It is accepted if the largest nodal residual falls to at most $(1 - 0.01\,\alpha)$ times its value before the step ($\max|F| = 6.1656 \times 10^{-4}$ kg/s), otherwise α is halved.

Table 9. case05, iteration 8: line search.

Trial	α	P after trial	max |F| after trial (kg/s)	Required ≤ (kg/s)	Result
1	1.000000	M: 102753.4555	2.7176×10⁻⁶	6.1039×10⁻⁴	accepted

After the step the densities are refreshed at the new pressures and the residual is measured again: $\max|F| = 2.7176 \times 10^{-6}$ kg/s against a tolerance of 1.0000×10^{-4} kg/s — **converged**.

Check against theory. Converged flow 53.450 m³/s vs 53.452 analytic; fan pressure 1437.3 Pa vs 1428.6 Pa at 1.2 kg/m³. The fan pressure is 0.6 % higher because the air at the fan is denser (1.2074 kg/m³: the fan raises the pressure at M by 1428 Pa, and denser air), and a fan's pressure scales with density; the flow is the same, because the system resistance scales with density too and the two effects cancel in $Q = \sqrt{P_0/(k + R)}$.

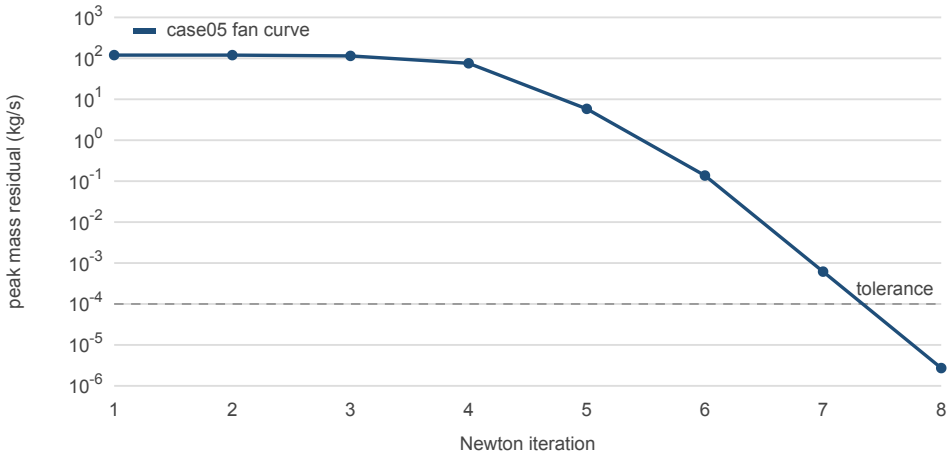


Figure 3. case05: peak mass residual after each iteration. Slow while the start-up zero flow is overcome, then quadratic.

A15. Example 3 — the balanced Wheatstone bridge (lit2)

The network of Figure 1. Resistances are chosen so that $R_{AB}/R_{BD} = R_{AC}/R_{CD} = 2/3$: the bridge is balanced, the pressures at B and C are equal, and the diagonal BC carries no air. Each side then carries $Q = \sqrt{\Delta P/(R_1 + R_2)}$: $\sqrt{300/0.5} = 24.495$ m³/s through B and $\sqrt{300/1.0} = 17.321$ m³/s through C (McPherson 1993, ch. 7). This case is a severe test of the solver: the true solution has an airway at exactly zero flow, the singular point of Section A10.

Table 10. lit2: every Newton iteration.

Iter.	P_B before (Pa)	P_C before (Pa)	max |F| before (kg/s)	α taken	max |F| after (kg/s)	Tolerance (kg/s)
1	101625.0000	101625.0000	37.969	0.001953	37.968	1.000×10⁻⁴
2	101625.0000	101625.0000	37.968	0.250000	37.854	1.000×10⁻⁴
3	101624.9984	101624.9984	37.854	1.000000	35.000	1.000×10⁻⁴
4	101623.8647	101623.8651	35.000	1.000000	22.091	1.000×10⁻⁴
5	101597.4306	101597.4303	22.091	1.000000	4.9343	1.000×10⁻⁴
6	101529.0032	101529.0035	4.9343	1.000000	0.14777	1.000×10⁻⁴
7	101505.7930	101505.7927	0.14777	1.000000	0.029374	1.000×10⁻⁴
8	101505.2138	101505.2141	0.029374	0.500000	8.7610×10⁻⁴	1.000×10⁻⁴
9	101505.2138	101505.2138	8.7610×10⁻⁴	1.000000	8.5898×10⁻⁴	1.000×10⁻⁴
10	101505.2137	101505.2137	8.5898×10⁻⁴	0.500000	6.4794×10⁻⁶	1.000×10⁻⁴

Iteration 1. B and C start at the intake pressure (all nodes are at the same elevation), so AB, AC and BC all have zero flow and the 10^6 slope cap. The capped slopes dominate the Jacobian, and the step is microscopic:

Starting point of iteration 1. The node pressures entering this iteration are listed below; the boundary pressures never change.

Table 11. lit2, iteration 1: node pressures before the step.

Node	Role	Pressure P (Pa)
A	fixed (boundary)	101625.0000
B	free (unknown)	101625.0000
C	free (unknown)	101625.0000
D	fixed (boundary)	101325.0000

Step 1: the flow in every airway. For each airway the effective pressure difference is $\Delta = P_s - P_t - \rho g\,(z_t - z_s)$, and the flow follows from the inverted square law $q = \operatorname{sign}(\Delta)\sqrt{|\Delta|/R}$. The slope $s = \partial q/\partial\Delta = 1/(2R|q|)$ is what the Jacobian needs.

Table 12. lit2, iteration 1: airway flows and slopes.

Airway	s→t	P_s − P_t (Pa)	ρgΔz (Pa)	Δ (Pa)	R (Ns²/m⁸)	q (m³/s)	ρ (kg/m³)	ρq (kg/s)	s (m³/s per Pa)
AB	A→B	0.0000	0.000	0.0000	0.20000	0.00000	1.20248	0.0000	10^6 (cap)
AC	A→C	0.0000	0.000	0.0000	0.40000	0.00000	1.20248	0.0000	10^6 (cap)
BD	B→D	300.0000	0.000	300.0000	0.30000	31.62278	1.20069	37.9693	0.052705
CD	C→D	300.0000	0.000	300.0000	0.60000	22.36068	1.20069	26.8483	0.037268
BC	B→C	0.0000	0.000	0.0000	0.50000	0.00000	1.20248	0.0000	10^6 (cap)

For example, airway **BD**: $\Delta = 101625.000 - 101325.000 - (0.000) = 300.0000$ Pa, so $q = \sqrt{300.0000/0.30000} = 31.62278$ m³/s, and $s = 1/(2 \times 0.30000 \times 31.6228) = 0.052705$.

Step 2: the mass residual at every free node. $F_i = \sum_{\text{in}} \rho_e q_e - \sum_{\text{out}} \rho_e q_e$ (kg/s), with "in" meaning airways whose target is node i.

$$F_B = 1.20248 \times (0.0000) - 1.20069 \times (31.6228) - 1.20248 \times (0.0000) = -37.96926\ \text{kg/s}$$

$$F_C = 1.20248 \times (0.0000) - 1.20069 \times (22.3607) + 1.20248 \times (0.0000) = -26.84832\ \text{kg/s}$$

Step 3: the Jacobian. Raising P_i by 1 Pa raises Δ on every airway leaving i and lowers it on every airway entering i, so $J_{ii} = -\sum_e \rho_e s_e$ over the airways at i, and $J_{ij} = +\rho_e s_e$ for an airway between i and j.

$$J_{BB} = -(1.20248 \times 1.0000 \times 10^6 + 1.20069 \times 0.052705 + 1.20248 \times 1.0000 \times 10^6) = -2.40$$

$$J_{CC} = -(1.20248 \times 1.0000 \times 10^6 + 1.20069 \times 0.037268 + 1.20248 \times 1.0000 \times 10^6) = -2.40$$

$$J_{BC} = J_{CB} = 1.20248 \times 10^6$$

Step 4: the Newton step. Solve $J\,\delta P = -F$.

$$\begin{bmatrix} -2.40495 \times 10^6 & 1.20248 \times 10^6 \\ 1.20248 \times 10^6 & -2.40495 \times 10^6 \end{bmatrix} \begin{bmatrix} \delta P_B \\ \delta P_C \end{bmatrix} = \begin{bmatrix} 37.9693 \\ 26.8483 \end{bmatrix}$$

By Cramer's rule, with determinant $\det J = J_{11}J_{22} - J_{12}J_{21} = 4.33785 \times 10^{12}$:

$$\delta P_B = \frac{(37.9693)(-2.40495 \times 10^6) - (1.20248 \times 10^6)(26.8483)}{4.33785 \times 10^{12}} = -2.84931 \times 10^{-5}\ \text{Pa}$$

$$\delta P_C = \frac{(-2.40495\times10^{6})(26.8483) - (1.20248\times10^{6})(37.9693)}{4.33785\times10^{12}} = -2.54103\times10^{-5}\ \mathrm{Pa}$$

Step 5: the line search. The full step is tried first. It is accepted if the largest nodal residual falls to at most $(1-0.01\,\alpha)$ times its value before the step ($\max|F| = 37.969$ kg/s), otherwise α is halved.

Table 13. lit2, iteration 1: line search.

Trial	α	P after trial	max |F| after trial (kg/s)	Required ≤ (kg/s)	Result
1	1.000000	B: 101625.0000 C: 101625.0000	37.952	37.590	rejected, halve α
2	0.500000	B: 101625.0000 C: 101625.0000	37.957	37.779	rejected, halve α
3	0.250000	B: 101625.0000 C: 101625.0000	37.961	37.874	rejected, halve α
4	0.125000	B: 101625.0000 C: 101625.0000	37.963	37.922	rejected, halve α
5	0.062500	B: 101625.0000 C: 101625.0000	37.965	37.946	rejected, halve α
6	0.031250	B: 101625.0000 C: 101625.0000	37.966	37.957	rejected, halve α
7	0.015625	B: 101625.0000 C: 101625.0000	37.967	37.963	rejected, halve α
8	0.007813	B: 101625.0000 C: 101625.0000	37.968	37.966	rejected, halve α
9	0.003906	B: 101625.0000 C: 101625.0000	37.968	37.968	rejected, halve α
10	0.001953	B: 101625.0000 C: 101625.0000	37.968	37.969	accepted

After the step the densities are refreshed at the new pressures and the residual is measured again: $\max|F| = 37.968$ kg/s against a tolerance of 1.0000×10^{-4} kg/s, so the solver continues.

Iterations 2–7. The slopes fall as flows develop (iteration 2: 4740 and 3548; iteration 3: 27.7 and 19.7), the Armijo test passes at α = 1 from iteration 3, and the residual drops 35.0 → 22.1 → 4.93 → 0.148 → 0.0294 kg/s. Iteration 5 in full:

Starting point of iteration 5. The node pressures entering this iteration are listed below; the boundary pressures never change.

Table 14. lit2, iteration 5: node pressures before the step.

Node	Role	Pressure P (Pa)
A	fixed (boundary)	101625.0000
B	free (unknown)	101597.4306
C	free (unknown)	101597.4303
D	fixed (boundary)	101325.0000

Step 1: the flow in every airway. For each airway the effective pressure difference is $\Delta = P_s - P_t - \rho g\,(z_t - z_s)$, and the flow follows from the inverted square law $q = \operatorname{sign}(\Delta)\sqrt{|\Delta|/R}$. The slope $s = \partial q/\partial\Delta = 1/(2R|q|)$ is what the Jacobian needs.

Table 15. lit2, iteration 5: airway flows and slopes.

Airway	s→t	P_s − P_t (Pa)	ρgΔz (Pa)	Δ (Pa)	R (Ns²/m⁸)	q (m³/s)	ρ (kg/m³)	ρq (kg/s)	s (m³/s per Pa)
AB	A→B	27.5694	0.000	27.5694	0.20000	11.74083	1.20231	14.1161	0.21293
AC	A→C	27.5697	0.000	27.5697	0.40000	8.30207	1.20231	9.9817	0.15056
BD	B→D	272.4306	0.000	272.4306	0.30000	30.13473	1.20053	36.1776	0.055307
CD	C→D	272.4303	0.000	272.4303	0.60000	21.30846	1.20053	25.5814	0.039108
BC	B→C	0.0003	0.000	0.0003	0.50000	0.02452	1.20215	0.0295	40.788

For example, airway **AB**: $\Delta = 101625.000 - 101597.431 - (0.000) = 27.5694$ Pa, so $q = \sqrt{27.5694/0.20000} = 11.74083$ m³/s, and $s = 1/(2 \times 0.20000 \times 11.7408) = 0.21293$.

Step 2: the mass residual at every free node. $F_i = \sum_{\text{in}} \rho_e q_e - \sum_{\text{out}} \rho_e q_e$ (kg/s), with "in" meaning airways whose target is node i.

$$F_B = 1.20231 \times (11.7408) - 1.20053 \times (30.1347) - 1.20215 \times (0.0245) = -22.09096\ \text{kg/s}$$

$$F_C = 1.20231 \times (8.3021) - 1.20053 \times (21.3085) + 1.20215 \times (0.0245) = -15.57029\ \text{kg/s}$$

Step 3: the Jacobian. Raising P_i by 1 Pa raises Δ on every airway leaving i and lowers it on every airway entering i, so $J_{ii} = -\sum_e \rho_e s_e$ over the airways at i, and $J_{ij} = +\rho_e s_e$ for an airway between i and j.

$$J_{BB} = -(1.20231 \times 0.21293 + 1.20053 \times 0.055307 + 1.20215 \times 40.788) = -49.3556$$

$$J_{CC} = -(1.20231 \times 0.15056 + 1.20053 \times 0.039108 + 1.20215 \times 40.788) = -49.2612$$

$$J_{BC} = J_{CB} = 49.0332$$

Step 4: the Newton step. Solve $J\,\delta P = -F$.

$$\begin{bmatrix} -49.3556 & 49.0332 \\ 49.0332 & -49.2612 \end{bmatrix} \begin{bmatrix} \delta P_B \\ \delta P_C \end{bmatrix} = \begin{bmatrix} 22.0910 \\ 15.5703 \end{bmatrix}$$

By Cramer's rule, with determinant $\det J = J_{11}J_{22} - J_{12}J_{21} = 27.0607$:

$$\delta P_B = \frac{(22.0910)(-49.2612) - (49.0332)(15.5703)}{27.0607} = -68.4273\ \text{Pa}$$

$$\delta P_C = \frac{(-49.3556)(15.5703) - (49.0332)(22.0910)}{27.0607} = -68.4267\ \text{Pa}$$

Step 5: the line search. The full step is tried first. It is accepted if the largest nodal residual falls to at most $(1 - 0.01\,\alpha)$ times its value before the step ($\max|F| = 22.091$ kg/s), otherwise α is halved.

Table 16. lit2, iteration 5: line search.

Trial	α	P after trial	max |F| after trial (kg/s)	Required ≤ (kg/s)	Result
1	1.000000	B: 101529.0032 C: 101529.0035	4.9359	21.870	accepted

After the step the densities are refreshed at the new pressures and the residual is measured again: $\max|F| = 4.9343$ kg/s against a tolerance of 1.0000×10^{-4} kg/s, so the solver continues.

Iterations 8–10: the diagonal near zero. Now the flow in BC is tiny (≈ 10^{-3} m³/s), its slope large (≈ 1400), and Section A10's mirror effect appears: the full step would flip the sign of the BC flow. In iteration 8 the line search rejects the full step and halves it; iteration 9 takes a full step that barely helps ($8.76 \times 10^{-4} \rightarrow 8.59 \times 10^{-4}$), and iteration 10 halves again and lands at 6.5×10^{-6} kg/s, below the 10^{-4} kg/s tolerance. The last iteration in full:

Starting point of iteration 10. The node pressures entering this iteration are listed below; the boundary pressures never change.

Table 17. lit2, iteration 10: node pressures before the step.

Node	Role	Pressure P (Pa)
A	fixed (boundary)	101625.0000
B	free (unknown)	101505.2137
C	free (unknown)	101505.2137
D	fixed (boundary)	101325.0000

Step 1: the flow in every airway. For each airway the effective pressure difference is $\Delta = P_s - P_t - \rho g\,(z_t - z_s)$, and the flow follows from the inverted square law $q = \mathrm{sign}(\Delta)\sqrt{|\Delta|/R}$. The slope $s = \partial q/\partial\Delta = 1/(2R|q|)$ is what the Jacobian needs.

Table 18. lit2, iteration 10: airway flows and slopes.

Airway	s→t	P_s − P_t (Pa)	ρgΔz (Pa)	Δ (Pa)	R (Ns²/m⁸)	q (m³/s)	ρ (kg/m³)	ρq (kg/s)	s (m³/s per Pa)
AB	A→B	119.7863	0.000	119.7863	0.20000	24.47308	1.20176	29.4109	0.10215
AC	A→C	119.7863	0.000	119.7863	0.40000	17.30508	1.20176	20.7966	0.072233
BD	B→D	180.2137	0.000	180.2137	0.30000	24.50943	1.19998	29.4109	0.068001
CD	C→D	180.2137	0.000	180.2137	0.60000	17.33079	1.19998	20.7966	0.048084
BC	B→C	2.56×10^{-7}	0.000	2.56×10^{-7}	0.50000	0.00072	1.20105	0.0009	1398.3

For example, airway **AB**: $\Delta = 101625.000 - 101505.214 - (0.000) = 119.7863$ Pa, so $q = \sqrt{119.7863/0.20000} = 24.47308$ m³/s, and $s = 1/(2 \times 0.20000 \times 24.4731) = 0.10215$.

Step 2: the mass residual at every free node. $F_i = \sum_{\text{in}} \rho_e q_e - \sum_{\text{out}} \rho_e q_e$ (kg/s), with "in" meaning airways whose target is node i.

$$F_B = 1.20176 \times (24.4731) - 1.19998 \times (24.5094) - 1.20105 \times (0.0007) = -0.00086\ \mathrm{kg/s}$$

$$F_C = 1.20176 \times (17.3051) - 1.19998 \times (17.3308) + 1.20105 \times (0.0007) = 0.00086\ \mathrm{kg/s}$$

Step 3: the Jacobian. Raising P_i by 1 Pa raises Δ on every airway leaving i and lowers it on every airway entering i, so $J_{ii} = -\sum_e \rho_e s_e$ over the airways at i, and $J_{ij} = +\rho_e s_e$ for an airway between i and j.

$$J_{BB} = -(1.20176 \times 0.10215 + 1.19998 \times 0.068001 + 1.20105 \times 1398.3) = -1679.59$$

$$J_{CC} = -(1.20176 \times 0.072233 + 1.19998 \times 0.048084 + 1.20105 \times 1398.3) = -1679.53$$

$$J_{BC} = J_{CB} = 1679.39$$

Step 4: the Newton step. Solve $J\,\delta P = -F$.

$$\begin{bmatrix} -1679.59 & 1679.39 \\ 1679.39 & -1679.53 \end{bmatrix} \begin{bmatrix} \delta P_B \\ \delta P_C \end{bmatrix} = \begin{bmatrix} 8.58981 \times 10^{-4} \\ -8.58981 \times 10^{-4} \end{bmatrix}$$

By Cramer's rule, with determinant $\det J = J_{11}J_{22} - J_{12}J_{21} = 585.920$:

$$\delta P_B = \frac{(8.58981 \times 10^{-4})(-1679.53) - (1679.39)(-8.58981 \times 10^{-4})}{585.920} = -2.11737 \times 10^{-7}\ \text{Pa}$$

$$\delta P_C = \frac{(-1679.59)(-8.58981 \times 10^{-4}) - (1679.39)(8.58981 \times 10^{-4})}{585.920} = 2.99721 \times 10^{-7}\ \text{Pa}$$

Step 5: the line search. The full step is tried first. It is accepted if the largest nodal residual falls to at most $(1 - 0.01\,\alpha)$ times its value before the step ($\max|F| = 8.5898 \times 10^{-4}$ kg/s), otherwise α is halved.

Table 19. lit2, iteration 10: line search.

Trial	α	P after trial	max |F| after trial (kg/s)	Required ≤ (kg/s)	Result
1	1.000000	B: 101505.2137 C: 101505.2137	8.5896×10⁻⁴	8.5039×10⁻⁴	rejected, halve α
2	0.500000	B: 101505.2137 C: 101505.2137	6.4794×10⁻⁶	8.5469×10⁻⁴	accepted

After the step the densities are refreshed at the new pressures and the residual is measured again: $\max|F| = 6.4794 \times 10^{-6}$ kg/s against a tolerance of 1.0000×10^{-4} kg/s — **converged**.

Check against theory. Flows $q_{AB} = 24.473$, $q_{BD} = 24.509$ (24.495 expected); $q_{AC} = 17.305$, $q_{CD} = 17.331$ (17.321 expected); $q_{BC} = 0.00072$ (0 expected). B and C are at 101 505.214 Pa, both $101\,625 - 0.2 \times 24.495^2 = 101\,505.0$ Pa by hand at standard density. The differences of a few hundredths of a m³/s between AB and BD are the density correction: air expands slightly from A to D (1.2018 → 1.2000 kg/m³), so the same mass is a larger volume downstream.

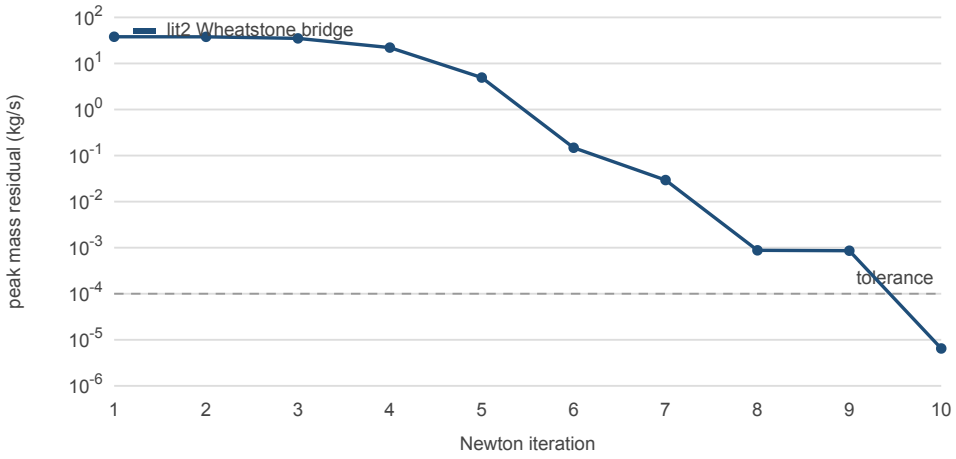


Figure 4. lit2: peak mass residual after each iteration.

A16. Example 4 — natural ventilation (case09)

A 500 m downcast shaft S1→B1 at 5 °C, a 400 m level B1→B2 and a 500 m upcast B2→S2 at 35 °C. Both shaft collars are at 101 325 Pa and there is no fan: only the density difference between the shafts drives air. B1 and B2 are free.

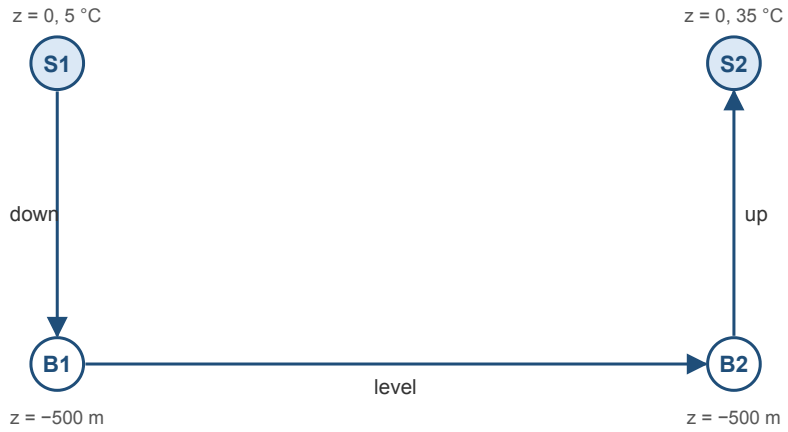


Figure 5. case09: natural ventilation through two shafts.

Initial estimate. The free nodes start at the collar pressure plus the weight of a 500 m column at the collar's density: $101\,325 + \rho_0 \times 9.79 \times 500 = 107527.1$ Pa, so $\rho_0 = 1.26703$ kg/m³ (5 °C air at 101 325 Pa, 50 % RH).

The hydrostatic terms. In the first iteration the downcast density is 1.30587 kg/m³ and the upcast 1.16860; the column terms are $\rho g\,\Delta z = 1.30587 \times 9.79 \times (-500) = -6392.3$ Pa down and $1.16860 \times 9.79 \times (+500) = 5720.3$ Pa up. Around the loop they leave 672.0 Pa uncancelled. That is the natural ventilation pressure, which the friction of the three airways must absorb.

Table 20. case09: every Newton iteration.

Iter.	P_{B1} before (Pa)	P_{B2} before (Pa)	max |F| before (kg/s)	α taken	max |F| after (kg/s)	Tolerance (kg/s)
1	107527.1312	107527.1312	573.06	1.000000	497.96	4.983×10^{-4}
2	107406.4455	107406.4451	497.96	1.000000	473.57	4.911×10^{-4}
3	107396.9629	107395.7280	473.57	1.000000	341.05	4.679×10^{-4}
4	107426.6222	107361.9325	341.05	1.000000	83.398	3.630×10^{-4}
5	107544.7530	107230.1341	83.398	1.000000	0.94867	3.175×10^{-4}
6	107586.7929	107183.7470	0.94867	1.000000	0.018189	3.169×10^{-4}
7	107587.2888	107183.3424	0.018189	1.000000	3.7709×10^{-4}	3.169×10^{-4}
8	107587.2990	107183.3372	3.7709×10^{-4}	1.000000	8.3963×10^{-6}	3.169×10^{-4}

Iteration 1 in full (note the 10^6 cap on the level, which starts with B1 = B2):

Starting point of iteration 1. The node pressures entering this iteration are listed below; the boundary pressures never change.

Table 21. case09, iteration 1: node pressures before the step.

Node	Role	Pressure P (Pa)
S1	fixed (boundary)	101325.0000
B1	free (unknown)	107527.1312
B2	free (unknown)	107527.1312
S2	fixed (boundary)	101325.0000

Step 1: the flow in every airway. For each airway the effective pressure difference is $\Delta = P_s - P_t - \rho g\,(z_t - z_s)$, and the flow follows from the inverted square law $q = \mathrm{sign}(\Delta)\sqrt{|\Delta|/R}$. The slope $s = \partial q/\partial\Delta = 1/(2R|q|)$ is what the Jacobian needs.

Table 22. case09, iteration 1: airway flows and slopes.

Airway	s→t	P_s − P_t (Pa)	ρgΔz (Pa)	Δ (Pa)	R (Ns²/m⁸)	q (m³/s)	ρ (kg/m³)	ρq (kg/s)	s (m³/s per Pa)
down	S1→B1	−6202.1312	−6392.258	190.1263	2.2392×10^{-3}	291.39296	1.30587	380.5227	0.76631
level	B1→B2	0.0000	0.000	0.0000	6.5238×10^{-3}	0.00000	1.27419	0.0000	10^6 (cap)
up	B2→S2	6202.1312	5720.276	481.8551	2.0038×10^{-3}	490.38236	1.16860	573.0587	0.50885

For example, airway **down**: $\Delta = 101325.000 - 107527.131 - (-6392.258) = 190.1263$ Pa, so $q = \sqrt{190.1263/2.2392\times10^{-3}} = 291.39296$ m³/s, and $s = 1/(2\times2.2392\times10^{-3}\times291.3930) = 0.76631$.

Step 2: the mass residual at every free node. $F_i = \sum_{\text{in}}\rho_e q_e - \sum_{\text{out}}\rho_e q_e$ (kg/s), with "in" meaning airways whose target is node i.

$$F_{B1} = 1.30587\times(291.3930) - 1.27419\times(0.0000) = 380.52275\ \text{kg/s}$$

$$F_{B2} = 1.27419\times(0.0000) - 1.16860\times(490.3824) = -573.05873\ \text{kg/s}$$

Step 3: the Jacobian. Raising P_i by 1 Pa raises Δ on every airway leaving i and lowers it on every airway entering i, so $J_{ii} = -\sum_e \rho_e s_e$ over the airways at i, and $J_{ij} = +\rho_e s_e$ for an airway between i and j.

$$J_{B1B1} = -(1.30587 \times 0.76631 + 1.27419 \times 1.0000 \times 10^6) = -1.27419 \times 10^6$$

$$J_{B2B2} = -(1.27419 \times 1.0000 \times 10^6 + 1.16860 \times 0.50885) = -1.27419 \times 10^6$$

$$J_{B1B2} = J_{B2B1} = 1.27419 \times 10^6$$

Step 4: the Newton step. Solve $J\,\delta P = -F$.

$$\begin{bmatrix} -1.27419 \times 10^6 & 1.27419 \times 10^6 \\ 1.27419 \times 10^6 & -1.27419 \times 10^6 \end{bmatrix} \begin{bmatrix} \delta P_{B1} \\ \delta P_{B2} \end{bmatrix} = \begin{bmatrix} -380.523 \\ 573.059 \end{bmatrix}$$

By Cramer's rule, with determinant $\det J = J_{11}J_{22} - J_{12}J_{21} = 2.03277 \times 10^6$:

$$\delta P_{B1} = \frac{(-380.523)(-1.27419 \times 10^6) - (1.27419 \times 10^6)(573.059)}{2.03277 \times 10^6} = -120.686 \text{ Pa}$$

$$\delta P_{B2} = \frac{(-1.27419 \times 10^6)(573.059) - (1.27419 \times 10^6)(-380.523)}{2.03277 \times 10^6} = -120.686 \text{ Pa}$$

Step 5: the line search. The full step is tried first. It is accepted if the largest nodal residual falls to at most $(1 - 0.01\,\alpha)$ times its value before the step ($\max|F| = 573.06$ kg/s), otherwise α is halved.

Table 23. case09, iteration 1: line search.

Trial	α	P after trial	max |F| after trial (kg/s)	Required ≤ (kg/s)	Result
1	1.000000	B1: 107406.4455 B2: 107406.4451	495.82	567.33	accepted

After the step the densities are refreshed at the new pressures and the residual is measured again: $\max|F| = 497.96$ kg/s against a tolerance of 4.9827×10^{-4} kg/s, so the solver continues.

From iteration 2 every step is a full Newton step, and the residual falls 474 → 341 → 83.4 → 0.949 → 0.0182 → 3.8 × 10⁻⁴ → 8.4 × 10⁻⁶ kg/s. The last iteration in full:

Starting point of iteration 8. The node pressures entering this iteration are listed below; the boundary pressures never change.

Table 24. case09, iteration 8: node pressures before the step.

Node	Role	Pressure P (Pa)
S1	fixed (boundary)	101325.0000
B1	free (unknown)	107587.2990
B2	free (unknown)	107183.3372
S2	fixed (boundary)	101325.0000

Step 1: the flow in every airway. For each airway the effective pressure difference is $\Delta = P_s - P_t - \rho g\,(z_t - z_s)$, and the flow follows from the inverted square law $q = \operatorname{sign}(\Delta)\sqrt{|\Delta|/R}$. The slope $s = \partial q/\partial \Delta = 1/(2R|q|)$ is what the Jacobian needs.

Table 25. case09, iteration 8: airway flows and slopes.

Airway	s→t	P_s − P_t (Pa)	ρgΔz (Pa)	Δ (Pa)	R (Ns²/m⁸)	q (m³/s)	ρ (kg/m³)	ρq (kg/s)	s (m³/s per Pa)
down	S1→B1	−6262.2990	−6394.102	131.8030	2.2398×10⁻³	242.58167	1.30625	316.8727	0.92024
level	B1→B2	403.9617	0.000	403.9617	6.5158×10⁻³	248.99231	1.27262	316.8723	0.30819
up	B2→S2	5858.3372	5710.763	147.5741	2.0004×10⁻³	271.60832	1.16665	316.8725	0.92024

For example, airway **down**: $\Delta = 101325.000 - 107587.299 - (-6394.102) = 131.8030$ Pa, so $q = \sqrt{131.8030/2.2398\times10^{-3}} = 242.58167$ m³/s, and $s = 1/(2\times2.2398\times10^{-3}\times242.5817) = 0.92024$.

Step 2: the mass residual at every free node. $F_i = \sum_{\text{in}} \rho_e q_e - \sum_{\text{out}} \rho_e q_e$ (kg/s), with “in” meaning airways whose target is node i.

$$F_{B1} = 1.30625\times(242.5817) - 1.27262\times(248.9923) = 0.00038\ \text{kg/s}$$

$$F_{B2} = 1.27262\times(248.9923) - 1.16665\times(271.6083) = -0.00014\ \text{kg/s}$$

Step 3: the Jacobian. Raising P_i by 1 Pa raises Δ on every airway leaving i and lowers it on every airway entering i, so $J_{ii} = -\sum_e \rho_e s_e$ over the airways at i, and $J_{ij} = +\rho_e s_e$ for an airway between i and j.

$$J_{B1B1} = -(1.30625\times0.92024 + 1.27262\times0.30819) = -1.59428$$

$$J_{B2B2} = -(1.27262\times0.30819 + 1.16665\times0.92024) = -1.46581$$

$$J_{B1B2} = J_{B2B1} = 0.392206$$

Step 4: the Newton step. Solve $J\,\delta P = -F$.

$$\begin{bmatrix} -1.59428 & 0.392206 \\ 0.392206 & -1.46581 \end{bmatrix} \begin{bmatrix} \delta P_{B1} \\ \delta P_{B2} \end{bmatrix} = \begin{bmatrix} -3.77091\times10^{-4} \\ 1.44348\times10^{-4} \end{bmatrix}$$

By Cramer’s rule, with determinant $\det J = J_{11}J_{22} - J_{12}J_{21} = 2.18308$:

$$\delta P_{B1} = \frac{(-3.77091\times10^{-4})(-1.46581) - (0.392206)(1.44348\times10^{-4})}{2.18308} = 2.27261\times10^{-4}\ \text{Pa}$$

$$\delta P_{B2} = \frac{(-1.59428)(1.44348\times10^{-4}) - (0.392206)(-3.77091\times10^{-4})}{2.18308} = -3.76682\times10^{-5}\ \text{Pa}$$

Step 5: the line search. The full step is tried first. It is accepted if the largest nodal residual falls to at most $(1 - 0.01\,\alpha)$ times its value before the step ($\max|F| = 3.7709\times10^{-4}$ kg/s), otherwise α is halved.

Table 26. case09, iteration 8: line search.

Trial	α	P after trial	max |F| after trial (kg/s)	Required ≤ (kg/s)	Result
1	1.000000	B1: 107587.2992 B2: 107183.3372	9.3053×10^{-11}	3.7332×10^{-4}	accepted

After the step the densities are refreshed at the new pressures and the residual is measured again: $\max|F| = 8.3963 \times 10^{-6}$ kg/s against a tolerance of 3.1687×10^{-4} kg/s — **converged**.

Check: the loop balance. The NVP at the converged state is 683.3 Pa. The friction losses are Rq^2: $2.2398 \times 10^{-3} \times 242.58^2 + 6.5158 \times 10^{-3} \times 248.99^2 + 2.0004 \times 10^{-3} \times 271.61^2 = 683.3$ Pa. The two agree, as Kirchhoff's second law requires. The volume flows differ (242.6, 249.0, 271.6 m³/s) while the mass flow is the same in all three (316.87, 316.87, 316.87 kg/s): the warm upcast air is lighter, so the same mass takes more volume. The solver reports a 17 % density spread here and recommends a thermal solve. In this test the temperatures are imposed at the nodes; Part C computes them instead.

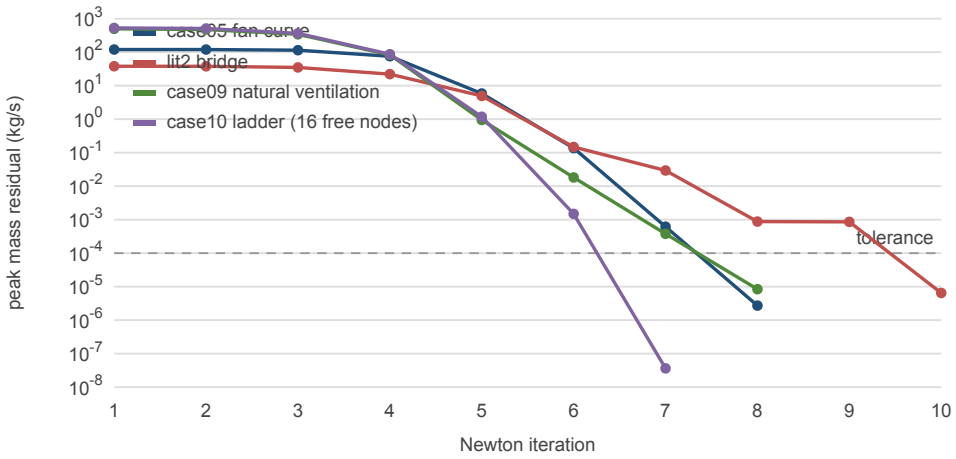


Figure 6. Residual histories of all four examples and the 16-node ladder.

A17. Larger networks

Nothing changes for a large mine except the size of the linear system in step 4. The ladder test case10 (two parallel drives joined by 8 crosscuts, 16 free nodes) converges in 7 iterations: 530 → 510 → 365 → 86.8 → 1.18 → 1.49×10^{-3} → 3.64×10^{-8} kg/s. The Palabora project (4,202 nodes, 4,477 airways; 2,770 free nodes after dead ends are pruned) converges in 39 iterations, each solving a 2,770-unknown system by the method of Part B. Its history (last figure of this part) shows both regimes: large full or half steps bring the residual from 1400 kg/s to about 0.01 kg/s in 19 iterations, after which the line search keeps cutting the step (down to 0.03125) and the residual falls only linearly to the tolerance (8.65×10^{-4} kg/s). That slow tail is the near-zero-flow effect of Section A10, spread over the many crosscuts and old workings of a real mine that carry almost no air. The work per iteration is dominated by the linear solve; the number of iterations depends mostly on how far the start is from the solution and on near-zero-flow airways, hardly at all on network size.

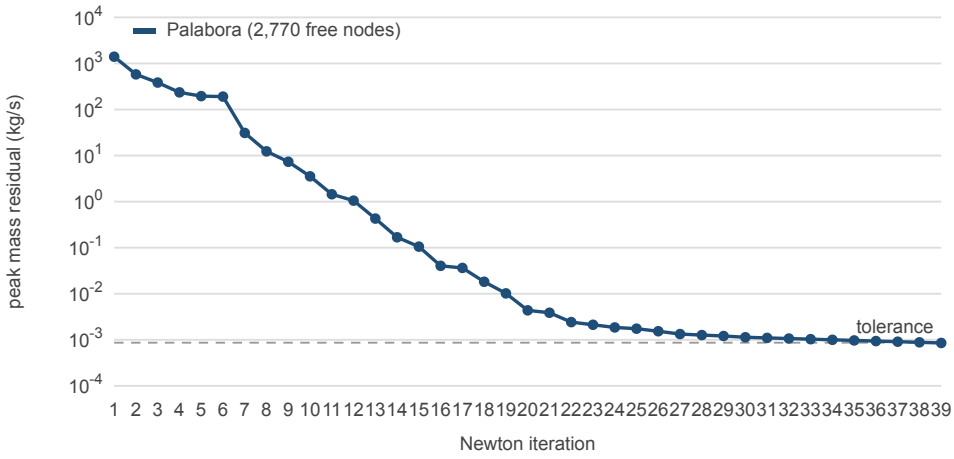


Figure 7. Palabora: peak mass residual per iteration.

Part B — Solving the linear systems

Every Newton iteration of Part A, and of the coupled solver in Part C, ends in a linear system $A\,x = b$. For a large mine this is where nearly all the computing time goes, so the choice of method matters. This part explains the methods the program uses: the **Cholesky factorisation** (direct, for the symmetric airflow system), stored in an **envelope** and made small by **reverse Cuthill–McKee ordering**; the **conjugate gradient** method with an **incomplete Cholesky** preconditioner as a fall-back for very large systems; and, for the non-symmetric coupled system, an **LU factorisation** polished by **GMRES**.

B1. The airflow system

The Newton step solves $J\,\delta P = -F$. Section A9 showed that J is symmetric with negative diagonal and nonnegative off-diagonals, weakly diagonally dominant, and strictly so in any row next to a boundary. The program multiplies by −1 and solves

$$B\,x = c, \qquad B = -J, \quad c = F, \quad x = \delta P \tag{B1}$$

B is then a **symmetric positive definite (SPD) M-matrix**: $x^{\mathsf{T}}Bx > 0$ for every $x \neq 0$. (Physically: raising a pattern of node pressures always pushes net air out of the region raised.) For SPD matrices the Cholesky factorisation exists, needs no pivoting, and is numerically stable.

B2. Gaussian elimination and the LU factorisation

Gaussian elimination subtracts multiples of row k from the rows below it to clear column k, for $k = 1, \dots, n-1$. Recording the multipliers gives the factorisation

$$A = L\,U \tag{B2}$$

with L lower triangular (unit diagonal, the multipliers $l_{ik} = a_{ik}^{(k)}/a_{kk}^{(k)}$ below it) and U upper triangular (the eliminated matrix). Then $Ax = b$ is solved in two triangular sweeps: *forward substitution* $Ly = b$, then *back substitution* $Ux = y$. Factorisation costs about $\tfrac{2}{3}n^3$ operations for a dense matrix, each solve $2n^2$. A zero or tiny pivot $a_{kk}^{(k)}$ breaks the method; general matrices need row exchanges (pivoting). SPD matrices never do.

B3. The Cholesky factorisation

For SPD B the factorisation can be made symmetric: $B = LL^{\mathsf{T}}$ with L lower triangular with a positive diagonal. Comparing entries of LL^{T} with B column by column gives the algorithm directly:

$$l_{jj} = \sqrt{b_{jj} - \sum_{k<j} l_{jk}^2}\,, \qquad l_{ij} = \frac{1}{l_{jj}}\Big(b_{ij} - \sum_{k<j} l_{ik}\,l_{jk}\Big) \quad (i > j) \tag{B3}$$

The square root's argument is positive at every step exactly when B is SPD, so the factorisation also *tests* definiteness: if a nonpositive value appears, the program abandons the direct solve and falls back to the iterative method of Section B6. Cholesky needs half the work and half the storage of LU ($\frac{1}{3}n^3$ operations dense) and no pivoting. The solve is then

$$L\,y = c \quad (\text{forward}), \qquad L^{\mathsf{T}} x = y \quad (\text{backward}) \tag{B4}$$

Worked example: the bridge, iteration 5

From Section A15, iteration 5 of the bridge has (free nodes B, C)

$$B = -J = \begin{bmatrix} 49.3556 & -49.0332 \\ -49.0332 & 49.2612 \end{bmatrix}, \qquad c = F = \begin{bmatrix} -22.0910 \\ -15.5703 \end{bmatrix}$$

Column 1:

$$l_{11} = \sqrt{49.3556} = 7.02536, \qquad l_{21} = \frac{-49.0332}{7.02536} = -6.97946$$

Column 2:

$$l_{22} = \sqrt{49.2612 - (-6.97946)^2} = \sqrt{0.548279} = 0.740459$$

Forward substitution $Ly = c$:

$$y_1 = \frac{-22.0910}{7.02536} = -3.14446, \qquad y_2 = \frac{-15.5703 - (-6.97946)(-3.14446)}{0.740459} = -50.6672$$

Back substitution $L^{\mathsf{T}} x = y$:

$$x_2 = \frac{-50.6672}{0.740459} = -68.4267, \qquad x_1 = \frac{-3.14446 - (-6.97946)(-68.4267)}{7.02536} = -68.4273$$

So $\delta P_B = -68.4273$ Pa and $\delta P_C = -68.4267$ Pa, identical to the Cramer's-rule result of Section A15 and to the solver's own step (-68.4273, -68.4267).

A larger example: the 16-node ladder

The ladder (case10) has 16 free nodes: T1…T8 along one drive and B1…B8 along the other, joined by crosscuts. Each row of B has at most 4 nonzeros. The table below shows the top-left corner of B in iteration 4 (in the solver's node order T1, B1, T2, B2, …).

Table 27. case10, iteration 4: top-left 5×5 corner of B = −J (kg/s per Pa).

	T1	B1	T2	B2	T3
T1	52.513	−16.316	−25.452	0	0
B1	−16.316	44.543	0	−25.452	0
T2	−25.452	0	124.41	−38.657	−60.300
B2	0	−25.452	−38.657	124.41	0
T3	0	0	−60.300	0	294.81

The first two columns of L by Eq. B3:

$$l_{11} = \sqrt{52.513} = 7.2466, \quad l_{21} = \frac{-16.316}{7.2466} = -2.2515, \quad l_{31} = \frac{-25.452}{7.2466} = -3.5123$$

$$l_{22} = \sqrt{44.543 - (-2.2515)^2} = 6.2828, \quad l_{32} = \frac{0 - (-3.5123)(-2.2515)}{6.2828} = -1.2587$$

Notice l_{32}: b_{32} is zero (T2 and B1 are not connected), yet l_{32} is not. Entries of L that are nonzero where B is zero are called **fill-in**. In this factor there are 7 of them (out of 120 positions below the diagonal). Continuing through all 16 columns and solving gives the Newton step to within 1.68×10^{-11} Pa of the solver's own.

B4. Sparsity, fill-in and the envelope

A mine network's matrix is extremely sparse: in the Palabora model 2,770 free nodes give 8,784 nonzeros, 3.17 per row, against 7,672,900 entries in the full matrix. A dense factorisation would ignore this and need about $\frac{1}{3}n^3 \approx 7.1 \times 10^9$ operations. But the factor of a sparse matrix is not as sparse as the matrix: elimination creates fill-in. The trick is to keep the fill-in close to the diagonal.

Envelope (profile, skyline) storage. For row i let f_i be the column of its first nonzero. A key property of Cholesky (and of LU without pivoting) is that L has no nonzeros to the left of f_i in row i: fill-in appears only *inside* the envelope $\{(i,j) : f_i \le j \le i\}$. The program therefore stores each row of L from f_i to the diagonal as one contiguous array and factorises inside it, skipping everything outside. The cost depends on the envelope size, not on n^2.

Ordering. The envelope depends entirely on how the nodes are numbered: renumbering the unknowns (a symmetric permutation PBP^{T}) changes nothing about the solution but everything about the envelope. The program uses the **reverse Cuthill–McKee** (RCM) ordering:

1. Start from a node of lowest degree (fewest connections), at the periphery of the network.
2. Number its neighbours next, in order of increasing degree; then the neighbours of those, and so on: a breadth-first sweep, level by level, across the network.
3. Reverse the final order (the reversal is known to reduce fill-in further, George 1971).

Neighbouring nodes then get nearby numbers, so every row's nonzeros sit in a narrow band left of the diagonal. A mine network is close to planar (levels stacked by shafts and ramps), so the breadth-first levels stay small and the band stays narrow.

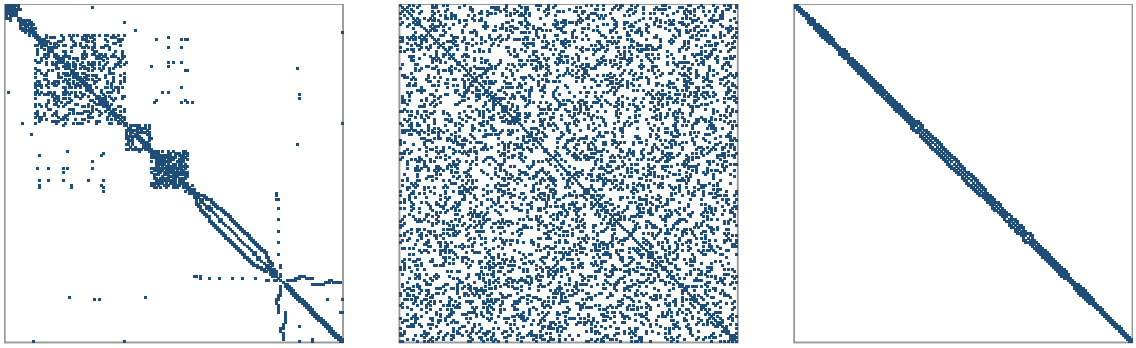


Figure 8. Nonzero pattern of the Palabora airflow matrix (2,770 × 2,770, each dot one block of about 17 × 17 entries) in three orders. Same matrix, same solution.

Table 28. Palabora: envelope size by node order.

Order	Bandwidth	Envelope (L + U entries)	Fraction of the full matrix
File order (as drawn)	2,520	375,298	4.89 %
Random order	2,758	3,979,560	51.87 %
Reverse Cuthill–McKee	42	136,378	1.78 %

RCM shrinks the envelope to 36 % of the file order and 3.4 % of a random order. The factorisation work grows roughly with the sum of the squared row lengths inside the envelope, so the gain in time is larger still. The program uses the direct solve whenever the envelope fits in 30 million entries (about 240 MB); beyond that it switches to the iterative method below.

B5. Why a direct solve

Earlier versions of the solver used the conjugate gradient method (next section) for every step. On real mine networks it struggled: airway conductances ρs range from about 4 to over 2×10^6 in the same matrix (a short wide drive next to a long narrow raise, or a near-zero-flow airway at the slope cap), and at that spread conjugate gradients lose accuracy to round-off. Measured on a 1,734-node model, 58 of 78 linear solves hit the 3,000-iteration cap with a true relative residual up to 13.5, worse than not solving at all. A direct factorisation has no such problem: its accuracy does not depend on convergence, and RCM makes it cheap.

B6. Conjugate gradients with incomplete Cholesky (fall-back)

For a network too large for the envelope, the program solves $Bx = c$ iteratively by the **preconditioned conjugate gradient** (PCG) method. CG minimises the energy norm of the error $\|x - x^*\|_B$ over a growing subspace, one search direction per iteration, with each new direction B-conjugate ($p_i^{\mathsf{T}} B p_j = 0$) to all previous ones. One iteration is

$$\begin{aligned} \alpha_k &= \frac{r_k^{\mathsf T} z_k}{p_k^{\mathsf T} B p_k}, \qquad x_{k+1} = x_k + \alpha_k p_k, \qquad r_{k+1} = r_k - \alpha_k B p_k, \\ z_{k+1} &= M^{-1} r_{k+1}, \quad p_{k+1} = z_{k+1} + \beta_k p_k, \qquad \beta_k = \frac{r_{k+1}^{\mathsf T} z_{k+1}}{r_k^{\mathsf T} z_k} \end{aligned} \tag{B5}$$

starting from $x_0 = 0$, $r_0 = c$, $z_0 = p_0 = M^{-1}c$. In exact arithmetic CG finishes in at most n iterations; in practice the iteration count depends on the condition number of $M^{-1}B$.

Worked example (no preconditioner, $M = I$). The same bridge system as Section B3. Iteration 1: $p_0 = c = \begin{bmatrix} -22.091 \\ -15.570 \end{bmatrix}$, $Bp_0 = \begin{bmatrix} -326.85 \\ 316.18 \end{bmatrix}$, so

$$\alpha_0 = \frac{c^{\mathsf T} c}{p_0^{\mathsf T} B p_0} = 0.317935, \quad x_1 = \alpha_0 p_0 = \begin{bmatrix} -7.02350 \\ -4.95034 \end{bmatrix}, \quad r_1 = \begin{bmatrix} 81.827 \\ -116.10 \end{bmatrix}, \quad \beta_0 = 27.618$$

Iteration 2: $p_1 = r_1 + \beta_0 p_0 = \begin{bmatrix} -528.29 \\ -546.12 \end{bmatrix}$, $\alpha_1 = 0.116231$, giving

$$x_2 = \begin{bmatrix} -68.4273 \\ -68.4267 \end{bmatrix}, \qquad r_2 = \begin{bmatrix} -4.97 \times 10^{-13} \\ 4.69 \times 10^{-13} \end{bmatrix}$$

the exact solution of Section B3 after $n = 2$ iterations, as the theory says. With thousands of unknowns, the point of a preconditioner is to get there in far fewer than n iterations.

Incomplete Cholesky IC(0). The preconditioner $M = \tilde L \tilde L^{\mathsf T}$ is a Cholesky factorisation that keeps only the entries where B itself is nonzero, discarding all fill-in. It is as sparse as B, cheap to apply (two triangular sweeps), and close enough to B to cut the iteration count many-fold. For an M-matrix IC(0) is guaranteed to exist (Meijerink & van der Vorst 1977); for degenerate user networks the program adds a small diagonal shift if needed. RCM ordering helps here too, because IC(0)'s quality depends on the order. The PCG stops at a relative residual of 10^{-6} or $\min(3000, 3n + 50)$ iterations.

B7. The coupled system is not symmetric

In the coupled airflow + thermal solver (Part C) the unknowns are, at every node, the pressure P, the enthalpy h and the moisture content r. The Jacobian couples them in one direction much more strongly than the other. The heat equation at a node depends on the enthalpy upstream, but not on the enthalpy downstream: air carries heat with it, it does not carry it back. The mass equation depends on enthalpy through density, but the heat equation depends on pressure only weakly. The result is a matrix with $J_{ij} \neq J_{ji}$. For example, from the first coupled iteration of case09 (Part C), at node B2:

$$\frac{\partial F^{\text{mass}}_{B2}}{\partial h_{B2}} = -8.042 \ \frac{\text{kg/s}}{\text{kJ/kg}}, \qquad \frac{\partial F^{\text{heat}}_{B2}}{\partial P_{B2}} = -1.613 \times 10^{-4} \ \frac{\text{kJ/kg}}{\text{Pa}}$$

CG requires a symmetric positive definite matrix, and Cholesky requires symmetry. Neither applies. The coupled system needs a method for general matrices.

B8. Why GMRES

The choice for a general sparse system is between a direct LU factorisation and a Krylov iterative method for non-symmetric matrices: GMRES, BiCGSTAB, QMR. The program combines the two:

1. **Envelope LU without pivoting**, in RCM order with each node's P, h and r numbered together. It reuses the envelope machinery of Section B4. Without pivoting the factorisation can meet a tiny pivot (a node no air reaches has a nearly empty mass row), so a pivot below 10^{-12} of its row's size is *nudged* to that size and counted. The factor is then the exact LU of a slightly different matrix.
2. **GMRES, preconditioned by that LU**, which removes the error the nudges introduced and certifies the accuracy of the answer by computing the true residual $\|b - Ax\|$.

Why GMRES rather than another Krylov method:

- **It minimises the residual.** At iteration k GMRES returns the vector in the Krylov space $\mathcal{K}_k = \mathrm{span}\{b, Ab, \ldots, A^{k-1}b\}$ with the smallest $\|b - Ax\|_2$. The residual can never increase from one iteration to the next. BiCGSTAB and CGS do not minimise anything and can stagnate or break down on exactly the badly scaled, strongly non-symmetric matrices the coupled problem produces (mass rows in kg/s next to enthalpy rows in kJ/kg).
- **It works for any nonsingular matrix**, with no symmetry or definiteness assumption.
- **Its cost is no concern here.** GMRES's weakness is that it stores one vector per iteration. With an exact or near-exact LU preconditioner it converges in 1–2 iterations, so the storage is a handful of vectors.
- **It verifies the answer.** A factorisation with nudged pivots could silently return a wrong Newton step. GMRES measures the true residual and keeps iterating until it is below 10^{-10}, so a bad step is never passed to the Newton loop.

In short: the LU does almost all the work, and GMRES is the safety net that turns “almost” into a guaranteed-accurate solve.

B9. How GMRES works

GMRES builds an orthonormal basis $v_1, \ldots, v_k$ of the Krylov space by the **Arnoldi process**, a Gram–Schmidt orthogonalisation of Av_j against the previous vectors:

$$v_1 = \frac{b}{\beta},\ \beta = \|b\|; \qquad w = Av_j, \quad h_{ij} = w^{\mathsf{T}} v_i, \quad w \leftarrow w - h_{ij} v_i \ (i \le j), \quad h_{j+1,j} = \|w\|, \tag{B8}$$

The coefficients form an upper Hessenberg matrix $\bar{H}_k$ ($(k+1) \times k$) with $AV_k = V_{k+1}\bar{H}_k$. Writing $x = V_k y$, the residual norm becomes $\|b - AV_k y\| = \|\beta e_1 - \bar{H}_k y\|$, a small least-squares problem. It is solved incrementally with Givens rotations, which zero the subdiagonal one entry at a time; the last entry of the rotated right-hand side, $|g_{k+1}|$, is the residual norm, available at no cost each iteration. **Right preconditioning** applies GMRES to

AM^{-1} (here M^{-1} is the LU solve) and recovers $x = M^{-1}V_k y$, so the residual it minimises is still the true one.

Worked example: a 3 × 3 block

Take the 3 × 3 block of the coupled Jacobian belonging to node B2 (unknowns P_{B2}, h_{B2}, r_{B2}; rows: mass, heat, moisture), with the right-hand side $-F$ from the same iteration:

$$A = \begin{bmatrix} -1.0470 & -8.0418 & 18.874 \\ -1.6130\times10^{-4} & 0.99993 & 1.5850\times10^{-4} \\ -7.1215\times10^{-6} & -3.5410\times10^{-6} & 1.0000 \end{bmatrix}, \qquad b = \begin{bmatrix} -901.94 \\ 0.20963 \\ 9.0143\times10^{-3} \end{bmatrix}$$

Iteration 1. $\beta = \|b\| = 901.936$, $v_1 = b/\beta = \begin{bmatrix} -1.000000 \\ 2.32417\times10^{-4} \\ 9.99440\times10^{-6} \end{bmatrix}$. Then $w = Av_1$, $h_{11} = w^{\mathsf T}v_1 = -1.04528$, and after removing that component $h_{21} = \|w\| = 6.37240\times10^{-4}$. The least-squares problem $\min_y \|\beta e_1 - \bar H_1 y\|$ with $\bar H_1 = (h_{11}, h_{21})^{\mathsf T}$ gives $y_1 = \beta h_{11}/(h_{11}^2 + h_{21}^2) = -862.868$, $x_1 = y_1 v_1$, and a residual $\beta\, h_{21}/\sqrt{h_{11}^2 + h_{21}^2} = 0.54985$, i.e. 0.06 % of $\|b\|$.

Table 29. GMRES on the B2 block, no preconditioner.

k	h_{1k} … $h_{k+1,k}$ (Arnoldi column)	Residual $\|b - Ax_k\| / \|b\|$	x_k
1	−1.0453, 6.3724×10^{-4}	6.096×10^{-4}	862.87, −0.20055, -8.6238×10^{-3}
2	7.2184, 0.99826, 6.5692×10^{-8}	3.994×10^{-11}	859.08, 0.34822, 0.015133
3	19.204, -4.6290×10^{-3}, 1.0000, 5.8245×10^{-6}	3.080×10^{-19}	859.08, 0.34822, 0.015133

After three iterations the Krylov space is all of $\mathbb{R}^3$ and the solution is exact: GMRES, like CG, terminates in at most n steps in exact arithmetic. The residual falls monotonically, as the minimisation property guarantees.

The full coupled system of case09

The whole first coupled iteration of case09 has 10 unknowns (P at the two free nodes, h and r at all four nodes). The table below shows GMRES on it with no preconditioner, with the diagonal (Jacobi) preconditioner, with the exact LU, and with the LU of a slightly wrong matrix (every diagonal entry 2 % too large, standing in for nudged pivots):

Table 30. Relative residual ‖b − Ax‖/‖b‖ after each GMRES iteration, case09 coupled system, iteration 1.

Iteration	No preconditioner	Jacobi	Exact LU	LU, 2 % perturbed
1	0.515	0.254	3.16×10^{-17}	3.69×10^{-3}
2	0.121	4.40×10^{-3}	—	9.09×10^{-6}
3	0.0368	1.04×10^{-3}	—	2.49×10^{-9}
4	6.37×10^{-5}	1.06×10^{-4}	—	2.06×10^{-11}
5	3.73×10^{-6}	4.53×10^{-6}	—	1.20×10^{-14}
6	4.04×10^{-16}	4.29×10^{-10}	—	—
7	—	1.26×10^{-16}	—	—

Without preconditioning GMRES needs 6 of the possible 10 iterations, because the rows have wildly different scales and the Krylov space must grow almost to the full dimension. The diagonal (Jacobi) preconditioner does not help here (7 iterations): it rescales the rows but cannot undo the strong one-way coupling between them. The exact LU preconditioner makes $AM^{-1} = I$, and GMRES finishes in one iteration at machine precision. The slightly wrong LU reaches the program's stopping test (10^{-10}) in 4 iterations. This is the regime the program runs in: on every coupled iteration of case09 GMRES took 1 iteration, with true relative residuals of 10^{-15}–10^{-17}.

For larger networks the same holds: GMRES restarts every 20 iterations and stops after 40 or at 10^{-10}, but with the envelope LU as preconditioner it is rarely asked for more than two.

Part C — The coupled airflow and thermal solver

C1. Why airflow and heat must be solved together

Part A treated the air temperature at every node as known. In a deep or hot mine it is not: the air warms as it descends (autocompression), takes heat and moisture from the rock, from machines and from people, and is cooled by coolers. The temperature, in turn, sets the density, and the density acts back on the airflow in four ways:

- resistance scales with density, $R = R_{\mathrm{std}}\,\rho/\rho_{\mathrm{ref}}$;
- the weight of every air column, $\rho g\,\Delta z$, and with it the natural ventilation pressure;
- the fan's pressure at its operating density;
- mass flow ρq, which is what is conserved at the nodes.

The heat, for its part, depends on the airflow: the mass flow carries it, and the air velocity sets the heat-transfer coefficient at the rock wall. So the two problems form one system of equations.

Solving one after the other (Picard iteration)

The straightforward approach alternates: solve the airflow with the current temperatures, then the temperatures with the new flows, and repeat until nothing changes. This is Picard (fixed-point) iteration. It converges when each round changes the answer by less than the round before, i.e. when the gain of one round, $|\partial q_{\mathrm{new}}/\partial q_{\mathrm{old}}|$, is below 1. Where a fan dominates, the temperatures change the flows only slightly, the gain is small, and Picard works.

Where natural ventilation drives the air, it fails. In a loop driven only by the density difference,

$$q = \sqrt{\frac{\mathrm{NVP}}{R_{\mathrm{loop}}}}, \qquad \mathrm{NVP} = gH\,(\bar{\rho}_{\mathrm{down}} - \bar{\rho}_{\mathrm{up}}) \tag{C1}$$

so $\partial q/\partial\,\mathrm{NVP} = 1/(2\sqrt{\mathrm{NVP}\,R_{\mathrm{loop}}})$, which grows without bound as the NVP, and the flow, become small. Meanwhile the NVP depends on the temperatures, and the temperatures on the flow: slow air in a hot shaft approaches the rock temperature. Each Picard round therefore swings the flow by the full amount. The program's earlier Picard solver, even with Aitken relaxation, converged on 8 of the 12 sample mines and oscillated on the rest.

Solving everything at once (Newton)

The coupled solver puts all unknowns, the pressures *and* the air state at every node, into one Newton system with all cross-derivatives. Newton's method does not care how strongly the equations are coupled, only that the Jacobian is right. This part describes the thermal equations, how they join the airflow equations, and follows the coupled solver through case09 by hand.

C2. The state of the air

At each node the air is described by three numbers: its pressure P (Pa), its specific enthalpy h (kJ per kg of dry air) and its moisture content r (kg of water vapour per kg of dry air). Enthalpy and moisture are the quantities that mixing and heat transfer conserve; temperatures follow from them:

$$h = 1.005\,t_{db} + r\,(2501 + 1.8\,t_{db}) \tag{C2}$$

so $t_{db} = (h - 2501r)/(1.005 + 1.8r)$; the wet bulb is found from r, P and t_{db} by inverting the psychrometer equation A5; the density follows from Eq. A6. If r exceeds saturation $r_s = 0.622\,e_s/(P - e_s)$ at the air's temperature, the excess condenses as mist (fog), keeping the enthalpy: the air is then saturated at a slightly higher temperature.

C3. Heat along an airway

Along an airway the steady-flow energy equation per kilogram of air is

$$h_{\text{out}} = h_{\text{in}} + \frac{g\,(z_{\text{in}} - z_{\text{out}})}{1000} + \frac{\dot{Q}_{\text{added}}}{\dot{m}} \tag{C3}$$

with $\dot{Q}$ in kW and $\dot{m}$ in kg/s. The middle term is **autocompression**: air descending 1000 m gains 9.79 kJ/kg, about 9.7 °C of dry-bulb rise if nothing else happens (and loses it again rising). Friction has no separate term: the fan's work enters the air as pressure and friction turns it into heat further on, so the fan's shaft power is added as heat at the fan. Heat sources (diesel engines at their rated power, load factor and efficiency; electrical machines; blasting: 3.7 MJ/kg of explosive, 20 % of it released as heat over the flushing time) add to $\dot{Q}$; coolers take heat and water out.

Rock heat: the Gibson/Starfield method

The rock around an airway starts at the **virgin rock temperature**, which rises with depth:

$$\text{VRT}(z) = \text{VRT}_{\text{datum}} + \frac{(z_{\text{collar}} - z - d_{\text{datum}}) \cdot \text{gradient}}{100} \tag{C4}$$

(gradient in °C per 100 m). Once ventilated, the rock near the wall cools and the heat flow falls with time. Modelling the airway as a cylinder of radius $r_0 = \text{Per}/2\pi$, the heat flux per square metre of wall after ventilation time t is

$$q = U\,(\text{VRT} - t_{\text{air}}), \qquad U = \frac{G(\text{Fo}, \text{Bi})\,k_r}{r_0}, \qquad \text{Fo} = \frac{\alpha t}{r_0^2},\ \ \text{Bi} = \frac{h_f r_0}{k_r},\ \ \alpha = \frac{k_r}{\rho_r c_r} \tag{C5}$$

where k_r, ρ_r, c_r are the rock's conductivity, density and specific heat, and G is Gibson's dimensionless heat flux. Instead of a curve fit, the program computes G from the exact solution for radial conduction outside a cylinder with a convective wall (Carslaw & Jaeger §13.5), whose Laplace transform

$$\bar{G}(p) = \frac{\text{Bi}\,\sqrt{p}\,K_1(\sqrt{p})}{p\,\big(\sqrt{p}\,K_1(\sqrt{p}) + \text{Bi}\,K_0(\sqrt{p})\big)} \tag{C6}$$

is inverted numerically by the Gaver–Stehfest algorithm (10 terms). The air-side coefficient comes from the airway's own friction factor by the Reynolds–Colburn analogy, $h_f = 0.71^{-2/3}\, k\, c_p\, |v|$ (at least 2 W/m²K for still air).

Part of the wall is wet (wetness fraction w, default 10 %). The dry part gives sensible heat, integrated exactly over each segment of length L_s since the air approaches the rock temperature:

$$\dot{Q}_s = \dot{m}\, c_p\, (\mathrm{VRT} - t_{db}) \left(1 - e^{-(1-w)\, U\, \mathrm{Per}\, L_s / (\dot{m} c_p)}\right) \tag{C7}$$

The wet part evaporates water: its heat $\dot{Q}_l = w\, U\, \mathrm{Per}\, L_s\, (\mathrm{VRT} - t_{wb})$ (the wet surface sits near the wet bulb) goes into latent heat, raising the moisture by $\Delta r = \dot{Q}_l / (\dot{m}\, (2501 - 2.383\, t_{wb}))$. The latent heat is capped so the air cannot pass saturation at the rock temperature. Airways are marched in segments of at most 50 m (at most 40 segments) so the heat follows the changing air temperature.

Validation: the Bluebook's tunnel example

The Bluebook (du Plessis 2014, ch. 22, Example 2) works the heat flow into a 3 m diameter circular tunnel in rock with $k_r = 6$ W/m°C and $\alpha = 2.5 \times 10^{-6}$ m²/s, ventilated for 4 years, with a virgin rock temperature of 44 °C. Then $r_0 = 1.5$ m and $\mathrm{Fo} = 2.5 \times 10^{-6} \times 4 \times 3.156 \times 10^{7} / 1.5^2 = 140.3$. The heat flow per metre of tunnel is $q' = 2\pi k_r\, G\, (\mathrm{VRT} - t_{\mathrm{eq}})$. The book reads its values from Starfield's and Goch & Patterson's tables; ASM Lab computes G by Eq. C6:

Table 31. Rock heat: ASM Lab against the Bluebook's worked example. The same comparison runs in the program's test suite.

Case	h (W/m²K)	t_eq (°C)	Bi	G (ASM Lab)	Heat flow, ASM Lab (W/m)	Heat flow, Bluebook (W/m)
Wet surface	59.3	26.9	14.82	0.3211	207.0	207
Dry surface	14	32	3.50	0.3012	136.3	136
Dry, insulated	0.79	32	0.20	0.1250	56.5	57
Bi → ∞ (no film)	—	—	∞	0.3277	—	T = 0.327 (Goch & Patterson)

All three cases agree to the precision the book quotes. The psychrometric constants of Sections A4 and C2 are those of the Blackbook (PEE 410, ch. 5): e_s with 0.6105 kPa, $H'_w = 1.8\, t + 2501$ kJ/kg, $\gamma = 0.000644$, and the latent heat $2501 - 2.383\, t$ of the Bluebook (ch. 19).

C4. Mixing at nodes

At a junction the incoming streams mix. Their mass flows $\dot{m}_e$ add, and so do their enthalpy and moisture flows:

$$h_n = \frac{\sum_{e\to n} |\dot m_e|\, h_{\text{out},e} + M_{\text{ext}}\, h_{\text{ext}}}{\sum_{e\to n} |\dot m_e| + M_{\text{ext}}}, \qquad r_n = \text{same with } r \tag{C8}$$

where the sum is over the airways delivering air *into* n (whichever way they are drawn), and M_{ext} is outside air a boundary node draws in (the air leaving it minus the air arriving), at the node's stated dry and wet bulb or the site's surface conditions.

Still air. A node that no air reaches has no mixing equation: 0/0. The coupled solver adds a small tie to a “still air” state, the rock-temperature air where the rock heats the node's airways (saturated if the rock is wet), otherwise the mean of the neighbours:

$$h_n = \frac{\sum |\dot m_e|\, h_{\text{out},e} + M_{\text{ext}} h_{\text{ext}} + \tau\, h_{\text{still}}}{\sum |\dot m_e| + M_{\text{ext}} + \tau}, \qquad \tau = \frac{\varepsilon^2}{M+\varepsilon}, \quad \varepsilon = 10^{-7} \max_e |\dot m_e| \tag{C9}$$

with M the air arriving. Where air flows ($M \gg \varepsilon$) the tie is $\approx \varepsilon^2/M$, utterly negligible; where none arrives it is ε and decides the state. The tie also makes the equations continuous as a flow passes through zero, which Newton's method needs.

C5. The coupled equations

The unknowns are the pressure at every free node and the enthalpy and moisture at *every* node (boundary nodes included: their pressure is fixed, but the air there is not, because an exhaust receives whatever arrives). At each node:

$$\begin{aligned} F_n^{\text{mass}} &= \textstyle\sum_{\text{in}} \rho_e q_e - \sum_{\text{out}} \rho_e q_e = 0 && (\text{free nodes; kg/s}) \\ F_n^{\text{heat}} &= h_n - h_{\text{mix},n} = 0 && (\text{kJ/kg}) \\ F_n^{\text{moist}} &= 1000\,\big(r_n - r_{\text{fog}}(h_{\text{mix},n}, r_{\text{mix},n})\big) = 0 && (\text{g/kg}) \end{aligned} \tag{C10}$$

where $h_{\text{mix}}, r_{\text{mix}}$ are the right-hand sides of Eq. C9, computed from the air *delivered* by each inflowing airway, i.e. its upstream node's air marched along it by Section C3. The densities in the mass equation come from each node's own P, h, r. The airflow equations are exactly those of Part A, with the temperatures now unknowns rather than inputs. Moisture is carried in g/kg so the three rows have comparable sizes.

Tolerances. Mass as in Part A (10^{-6} of the largest mass flow, at least 10^{-4} kg/s), and 10^{-6} kJ/kg for enthalpy and 10^{-6} g/kg for moisture: tight enough that the energy balance of a whole mine closes to about a watt (800 kg/s × 10^{-6} kJ/kg = 0.8 W).

C6. The coupled Jacobian

The Jacobian has a 3 × 3 block for each pair of neighbouring nodes:

$$J_{nk} = \begin{bmatrix} \partial F_n^{\text{mass}}/\partial P_k & \partial F_n^{\text{mass}}/\partial h_k & \partial F_n^{\text{mass}}/\partial r_k \\ \partial F_n^{\text{heat}}/\partial P_k & \partial F_n^{\text{heat}}/\partial h_k & \partial F_n^{\text{heat}}/\partial r_k \\ \partial F_n^{\text{moist}}/\partial P_k & \partial F_n^{\text{moist}}/\partial h_k & \partial F_n^{\text{moist}}/\partial r_k \end{bmatrix} \tag{C11}$$

- **Mass rows, pressure columns:** analytically, exactly as in Part A (Eq. A15).

- **Mass rows, h and r columns:** how a node's air changes the density of its airways, hence their resistance, column weight and mass flow. This is the natural-ventilation coupling.
- **Heat and moisture rows:** $\partial F/\partial h_n = 1$ on the diagonal, $-\partial h_{\text{mix}}/\partial h_k$ for each upstream neighbour k (the share of k's air in the mix, modified by the march), and pressure columns through the mass flows that weight the mix.

Everything except the analytic block is computed by **local finite differences**: nudge one unknown at one node ($\delta h = 10^{-4}\max(1,|h|)$, $\delta r = 10^{-7}$, $\delta P = \max(10^{-4}, 10^{-8}P)$), recompute that node's density, its airways (flow and delivered air) and the equations at the node and its neighbours, and divide the changes by the nudge. Only the neighbourhood is recomputed, so building the Jacobian costs a few airway marches per node, not a full solve per unknown.

The matrix is not symmetric (Part B7): heat flows downstream only. The unknowns are numbered node by node in RCM order with each node's P, h, r together, so each 3 × 3 block sits on or near the diagonal and the envelope LU of Part B8 stays small, with GMRES to finish (Part B9).

C7. The coupled Newton loop

1. **Start** from the airflow-only solution and a thermal solve on it (Picard's first round), so the first Newton step starts close.
2. **Evaluate** every node's temperatures and density from (P, h, r), every airway's flow (Part A), mass flow and delivered air (Section C3), and all residuals (Eq. C10).
3. **Build** the Jacobian (Section C6) and **solve** $J\,\delta x = -F$ by envelope LU + GMRES.
4. **Line search** on the scaled merit function $\Phi = \sum_i (F_i/\text{tol}_i)^2$: accept α if $\Phi(x + \alpha\,\delta x) \le (1 - 10^{-4}\alpha)\,\Phi(x)$, else halve, up to 12 times (if none passes, the best trial is taken).
5. **Stop** when every residual is within its tolerance ($\max_i |F_i|/\text{tol}_i \le 1$). As in Part A, the residual stagnating within 10× of tolerance for 6 iterations is reported as the precision limit, and no progress for 15 iterations as a stall.

C8. Worked example — natural ventilation with real temperatures (case09)

The network of Section A16 (two 500 m shafts and a 400 m level), now with the heat computed rather than imposed. Only the intake S1 keeps its stated air (5 °C, saturated); the temperatures at B1, B2 and S2 are unknowns. The site is the program's “SA Bushveld (Platinum)” geothermal profile:

Table 32. case09: thermal site data.

Parameter	Value
VRT at datum	38 °C at 1500 m below the collar
Geothermal gradient	1.2 °C per 100 m
Rock k, ρ, c	2.8 W/m·K, 2750 kg/m³, 820 J/kg·K
Age of workings (default)	5 years
Wet fraction of wall (default)	10 %
Surface air	22 °C dry bulb, 15 °C wet bulb, 101.325 kPa

Rock temperatures. By Eq. C4, at the mid-depth of the shafts (250 m) $\mathrm{VRT} = 38 + (250 - 1500) \times 1.2/100 = 23.0$ °C, and on the level (500 m) $38 + (500 - 1500) \times 1.2/100 = 26.0$ °C.

Table 33. case09 coupled: every Newton iteration (residuals after the step).

Iter.	GMRES iters	GMRES rel. residual	Step α	Mass resid. (kg/s)	Heat resid. (kJ/kg)	Moisture resid. (g/kg)
1	1	1.3×10⁻¹⁶	0.500	37.20	7.091	0.9538
2	1	2.3×10⁻¹⁵	0.500	13.15	3.515	0.4783
3	1	2.3×10⁻¹⁵	1.000	29.63	1.432	0.06327
4	1	2.1×10⁻¹⁶	1.000	8.745	1.268	0.05844
5	1	6.2×10⁻¹⁶	1.000	0.8984	0.09175	4.286×10⁻³
6	1	1.5×10⁻¹⁶	1.000	0.09139	5.834×10⁻⁴	2.632×10⁻⁵
7	1	6.9×10⁻¹⁷	1.000	2.396×10⁻³	2.425×10⁻⁶	1.074×10⁻⁷
8	1	4.8×10⁻¹⁶	1.000	4.733×10⁻⁵	3.397×10⁻⁸	1.504×10⁻⁹

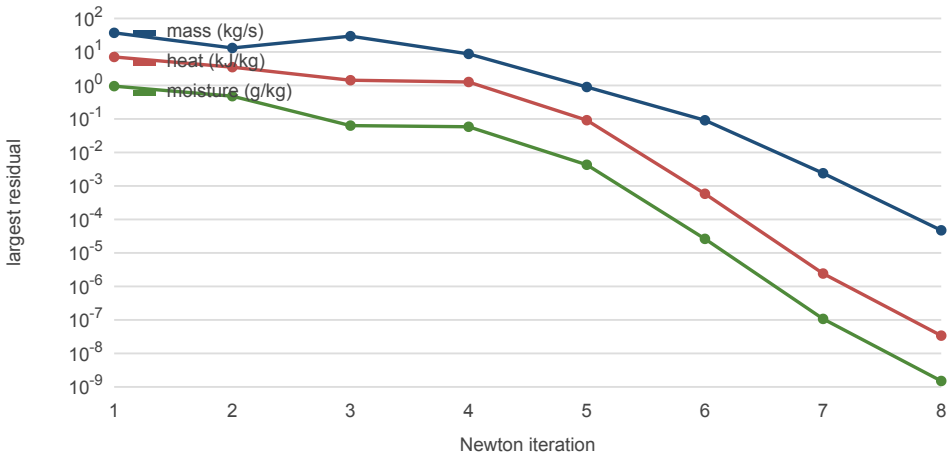


Figure 9. case09 coupled: the three residuals converge together, quadratically in the last iterations.

Iteration 1: the starting state

The starting state is the airflow-only solution followed by one thermal solve. Its flows were computed with the node temperatures stated in the file (5 °C down, 35 °C up), which drove 242.6 m³/s round the loop (Section A16). The thermal solve on those flows gives an upcast

far cooler than 35 °C, so the starting state is badly out of balance. The airway states at the start:

Table 34. case09 coupled, iteration 1: airway states at the start.

Airway	q (m³/s)	ṁ (kg/s)	Upstream → downstream	h delivered (kJ/kg)
down	168.47	217.69	S1 → B1	23.7787
level	244.80	322.30	B1 → B2	23.8463
up	−449.97	−579.64	S2 → B2	24.1768

With the cooler upcast, the start state's pressures no longer balance: the upcast now carries air *down* from S2 (negative flow), and both shafts feed B2, which has nowhere to send it. The residuals:

Table 35. case09 coupled, iteration 1: unknowns in solver order, residuals and the Newton step.

Unknown (column)	Equation (row)	Value now	Residual F	Tolerance	Newton step δx
h_{S2}	heat S2	19.1688 kJ/kg	−22.318	10^{-6}	22.318
r_{S2}	moist. S2	5.4244 g/kg	2.9627	10^{-6}	−2.9627
P_{B2}	mass B2	107183.337 Pa	901.94	5.80×10^{-4}	522.35
h_{B2}	heat B2	23.8491 kJ/kg	−0.20963	10^{-6}	14.572
r_{B2}	moist. B2	5.4152 g/kg	-9.0143×10^{-3}	10^{-6}	−1.8904
P_{B1}	mass B1	107587.299 Pa	−104.61	5.80×10^{-4}	49.449
h_{B1}	heat B1	23.6832 kJ/kg	−0.095455	10^{-6}	0.21650
r_{B1}	moist. B1	5.4078 g/kg	-4.1100×10^{-3}	10^{-6}	9.3276×10^{-3}
h_{S1}	heat S1	18.5756 kJ/kg	-2.7498×10^{-12}	10^{-6}	2.7498×10^{-12}
r_{S1}	moist. S1	5.3986 g/kg	-5.3169×10^{-13}	10^{-6}	5.3169×10^{-13}

The Jacobian (rows: equations, columns: unknowns, both in the solver's RCM order). Entries below 10^{-9} are shown as 0:

Table 36. case09 coupled, iteration 1: the 10 × 10 Jacobian.

	h_{S2}	r_{S2}	P_{B2}	h_{B2}	r_{B2}	P_{B1}	h_{B1}	r_{B1}	h_{S1}	r_{S1}
heat S2	1.00	0	0	0	0	0	0	0	0	0
moist. S2	0	1.00	0	0	0	0	0	0	0	0
mass B2	−7.58	17.8	−1.05	−8.04	18.9	0.399	−0.281	0.660	0	0
heat B2	−0.638	−0.0109	-1.61×10^{-4}	1.000	1.58×10^{-4}	1.66×10^{-4}	−0.354	-7.77×10^{-3}	0	0
moist. B2	9.57×10^{-5}	−0.643	-7.12×10^{-6}	-3.54×10^{-6}	1.00	7.29×10^{-6}	6.32×10^{-5}	−0.357	0	0
mass B1	0	0	0.399	0.280	−0.656	−2.13	−19.4	45.4	−19.2	51.2
heat B1	0	0	0	0	0	-2.34×10^{-3}	0.973	0.0643	−1.01	0.0269
moist. B1	0	0	0	0	0	-1.01×10^{-4}	-1.18×10^{-3}	1.00	-7.64×10^{-4}	−0.997
heat S1	0	0	0	0	0	0	0	0	1.00	0
moist. S1	0	0	0	0	0	0	0	0	0	1.00

Reading it: row “heat S2” has a 1 on its own diagonal and nothing else, because no airway delivers air into S2 at this moment (the upcast flows out of S2), so S2's air is simply the outside air it draws in. Row “mass B2” depends on every node's enthalpy: $\partial F_{B2}^{\text{mass}}/\partial h_{B2} = -8.042$ kg/s per kJ/kg says that warming the air at B2 by 1 kJ/kg (about 1 °C) lightens the air in both airways at B2, changing the column weights and resistances so that 8.0 kg/s more air leaves B2 than arrives. Row “heat B1” has $\partial F_{B1}^{\text{heat}}/\partial h_{S1} = -1.008$: B1's air is S1's air marched down the shaft, one for one.

The step is solved by envelope LU + GMRES in 1 GMRES iteration (relative residual 1.3×10^{-16}). The full step overshoots (the merit function rises), so the line search halves it: $\alpha = 0.5$. After the step the largest residuals are 37.20 kg/s, 7.091 kJ/kg and 0.9538 g/kg. Iterations 3–8 take full steps, and all three residuals fall quadratically together.

The converged state

Table 37. case09 coupled: converged node states.

Node	z (m)	P (Pa)	t_{db} (°C)	t_{wb} (°C)	r (g/kg)	h (kJ/kg)	ρ (kg/m³)
S1	0	101325.00	5.000	5.000	5.3986	18.5756	1.26497
B1	-500	107637.98	10.640	8.306	5.4389	24.4002	1.31704
B2	-500	107618.94	11.255	8.639	5.4716	25.1070	1.31394
S2	0	101325.00	7.197	6.255	5.5106	21.0866	1.25497

Table 38. case09 coupled: airways, compared with the airflow-only solve of Section A16.

Airway	q coupled (m³/s)	ṁ (kg/s)	q airflow-only (m³/s)	Autocompression (kW)	Rock sensible (kW)	Rock latent (kW)
down	54.18	69.953	242.58	342.42	58.03	7.00
level	53.18	69.953	248.99	0.00	43.77	5.67
up	54.46	69.953	271.61	−342.42	54.41	6.78

Checking the downcast shaft by hand

Air velocity and film coefficient. $v = q/A = 54.185/36 = 1.5051$ m/s; with the shaft's $k = 0.008$: $h_f = 0.71^{-2/3} \times 0.008 \times 1005 \times 1.5051 = 15.205$ W/m²K.

Gibson parameters. $r_0 = 24/2\pi = 3.8197$ m; $\alpha = 2.8/(2750 \times 820) = 1.2417 \times 10^{-6}$ m²/s; $\mathrm{Fo} = 1.2417 \times 10^{-6} \times 5 \times 3.156 \times 10^{7}/3.8197^2 = 13.428$; $\mathrm{Bi} = 15.205 \times 3.8197/2.8 = 20.743$. The Stehfest inversion of Eq. C6 gives $G = 0.4906$, so $U = 0.4906 \times 2.8/3.8197 = 0.3596$ W/m²K and, over the whole wall, $U\,\mathrm{Per}\,L = 0.3596 \times 24 \times 500 = 4316$ W/K.

Autocompression. $\dot{m}\,g\,\Delta z/1000 = 69.953 \times 9.79 \times 500/1000 = 342.42$ kW (program: 342.42 kW).

Rock heat, one-step estimate. With the mean air temperature in the shaft, $\bar{t}_{db} = (5.00 + 10.64)/2 = 7.82$ °C and $\bar{t}_{wb} = 6.65$ °C, against the mid-depth VRT of 23 °C:

$$\dot{Q}_s \approx 0.9 \times 4316 \times (23 - 7.82) = 58.96\,\mathrm{kW}, \qquad \dot{Q}_l \approx 0.1 \times 4316 \times (23 - 6.65) = 7.05\,\mathrm{kW}$$

against the program's 58.03 and 7.00 kW. The program's figures differ by a few per cent because it marches 10 segments of 50 m, each with its own rock temperature (20.3 °C at the top, 25.7 °C at the bottom) and the air temperature reached there.

Energy equation C3. $h_{B1} = 18.5756 + \frac{342.42+58.03+7.00}{69.953} = 18.5756 + 5.8246 = 24.4002$ kJ/kg; the solved enthalpy at B1 is 24.4002. Since only the downcast delivers air to B1, Eq. C8 makes h_{B1} equal to the air the shaft delivers.

Moisture. $\Delta r = \dot{Q}_l/(\dot{m}\,(2501 - 2.383\,\bar{t}_{wb})) = 6.997/(69.953 \times 2485.1) = 4.025 \times 10^{-5}$ kg/kg $= 0.0402$ g/kg, so $r_{B1} = 5.3986 + 0.0402 = 5.4389$ g/kg (solved: 5.4389).

Why the flow fell from 243 to 54 m³/s

The natural ventilation pressure. Mean densities: downcast $(1.26497 + 1.31704)/2 = 1.29101$, upcast $(1.31394 + 1.25497)/2 = 1.28445$ kg/m³. So $\mathrm{NVP} = 9.79 \times 500 \times (1.29101 - 1.28445) = 32.08$ Pa.

The friction it overcomes. $R_{\mathrm{down}} = R_{\mathrm{up}} = 0.008 \times 500 \times 24/6^6 = 2.058 \times 10^{-3}$ and $R_{\mathrm{level}} = 0.012 \times 400 \times 20/5^6 = 6.144 \times 10^{-3}$ N·s²/m⁸ at 1.2 kg/m³, scaled by density: $\sum Rq^2 = 2.2137 \times 10^{-3} \times 54.18^2 + 6.7353 \times 10^{-3} \times 53.18^2 + 2.2024 \times 10^{-3} \times 54.46^2 = 32.08$ Pa. The loop balances.

The airflow-only solve took the upcast at the 35 °C written on its nodes, an NVP of about 680 Pa and 243 m³/s. With the heat computed, the rock (20–26 °C) warms the air only to 11.3 °C at the bottom of the upcast, which cools again to 7.2 °C by expansion as it rises. The density difference, and the NVP, are about 20 times smaller, and the flow $\propto \sqrt{\mathrm{NVP}}$ about 4.5 times smaller. Imposed temperatures can make natural ventilation look far stronger than the rock and the air can sustain; the coupled solve gives the self-consistent answer.

The energy balance

All energy entering the network must leave it. Air enters at S1 and leaves at S2 at the same elevation, so the potential-energy terms cancel and $\dot{m}\,(h_{S2} - h_{S1}) = 69.953 \times (21.0866 - 18.5756) = 175.65$ kW. The rock adds 175.65 kW (sensible + latent, all three airways), and autocompression down and expansion up cancel (342.42 and −342.42 kW). The balance closes to 8.92\times10^{-5} kW; the program reports it to 10^{-12} kW, because it sums the energy flows exactly rather than from rounded node values.

C9. Status and limitations

The coupled solver is new and still under development. It converges on case09, as shown above, but of the 12 sample mines it currently converges on only 4. On the others the line search stalls, and on coal_seams the iterates diverge. For comparison, the earlier Picard loop (Section C1) converged on 8 of the 12. Until this is resolved the program keeps the airflow-only solve as the default and reports clearly when a coupled solve has not converged.

Table 39. Coupled solver on the sample mines (SA Bushveld site profile).

Sample	Size	Airflow only	Coupled
copper_porphyry	66 nodes, 113 airways	converged (0.1 s)	did not converge, 15 iter. (0.2 s)
ug2_reef	51 nodes, 59 airways	converged (0.0 s)	converged, 4 iter. (0.0 s)
merensky_reef	46 nodes, 63 airways	converged (0.0 s)	converged, 9 iter. (0.0 s)
coal_seams	660 nodes, 1216 airways	converged (0.2 s)	did not converge, 17 iter. (2.8 s)
gold_vein	142 nodes, 159 airways	converged (0.0 s)	did not converge, 15 iter. (0.2 s)
nickel_massive	24 nodes, 27 airways	converged (0.0 s)	did not converge, 15 iter. (0.0 s)
Untitled_Mine	7 nodes, 7 airways	converged (0.0 s)	converged, 2 iter. (0.0 s)
sample_test1	29 nodes, 35 airways	converged (0.0 s)	converged, 25 iter. (0.1 s)
sample_test2	11 nodes, 13 airways	converged (0.0 s)	did not converge, 20 iter. (0.0 s)
sample_test3	14 nodes, 16 airways	converged (0.0 s)	did not converge, 24 iter. (0.1 s)
sample_test4	17 nodes, 21 airways	converged (0.0 s)	did not converge, 33 iter. (0.4 s)
sample_test5	26 nodes, 28 airways	converged (0.0 s)	did not converge, 22 iter. (0.1 s)

The failures share one cause, and the balanced bridge (lit2) reproduces it with four nodes: an airway at (nearly) zero flow. Part A's airflow solver passes through that point thanks to its line search on the largest mass residual (Section A10). In the coupled system the zero-flow airway also switches which node is upstream, and therefore which node's air the heat equations draw on. The analytic pressure slope there (capped at 10^6) and the finite-

difference heat derivatives describe the same point very differently, so the Newton direction is poor and the merit function, dominated by the tight heat tolerances, accepts only tiny steps. Remedies under development include regularising the flow law very near zero flow in the coupled mode (a short linear segment through the origin instead of the vertical square-root tangent) and balancing the merit function between the mass and heat rows. Until then, the airflow-only solver of Part A followed by a thermal solve remains the robust choice for large networks.

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